# Hazard Analysis
# Squigglr

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Table 1: Revision History

| Date | Developer(s) | Change |
|---|---|---|
| 14 October 2026 | Rayan Nasrallah | Added critical assumptions CA-001 to CA-006 and hazards H-RAY-001 to H-RAY-007 (#91). |
| 15 October 2026 | Rayan Nasrallah | Added my reflection responses to the appendix and my generative AI use disclosure (#140). |
| 15 October 2026 | Ahmed Elzaria | Added drawing-input and guided-lesson FMEA hazards H-AHM-001 to H-AHM-007 (part of #135). |
| 16 October 2026 | Shamil Canbolat | Added security and AI-assisted refinement FMEA hazards H-SHA-001 to H-SHA-005, my AI use disclosure, and reflection questions (part of #82). |
| 16 October 2026 | Moustafa Moustafa | Added sensory, feedback, progress-view and refinement hazards H-MOU-001 to H-MOU-006 (#115). |
| 16 October 2026 | Muhammad Huzaifah | Added the FMEA method, H-MUH-001 to H-MUH-008 and the coverage table (#127). |
| 16 October 2026 | Muhammad Huzaifah | Wrote the front matter, the Introduction with the hazard definition and template changes, the losses and out-of-scope items, and the system boundary and components (#126). |
| 16 October 2026 | Muhammad Huzaifah | Listed the HA-derived requirements with their hazards in Section 6 and planned their phases in the Section 7 roadmap (#128). |
| 17 October 2026 | Moustafa Moustafa | Added my reflection responses to the appendix and my generative AI use disclosure (#117). |

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# 1 Introduction

This document is the Hazard Analysis (HA) of Squigglr, a website that children use on a touchscreen device with a stylus to practise drawing, and that supporting adults use to choose that practice and follow its records. The HA asks how Squigglr could contribute to harm, and turns the answers into safety requirements recorded in the Software Requirements Specification (SRS). Items from the SRS are cited by their identifiers (IDs) as plain text: goals G.1 to G.6, constraints CON-###, facts FACT-###, business rules BR-###, assumptions ASM-###, requirement sources SRC-###, personas PER-1 to PER-3, open issues OI-##, likely changes LC-### and use cases UC-###. The subsections below give the background, the purpose and scope of this document, the definition of hazard it uses, the changes made to the course template, and a roadmap of the document.

## 1.1 Background

Squigglr supports the visual-motor practice of kindergarten and primary school learners, and of older children who need continued support, including some children with autism or attention-deficit/hyperactivity disorder (ADHD) (SRC-001). Its core, set by goals G.1 to G.6, has four parts: guided drawing lessons whose stages give gradually less help, from a demonstration through tracing a full outline and drawing between endpoints to drawing with no guide; a creative space where Squigglr refines a drawing a little, keeps it editable and lets the child restore the original or, if the child cannot, an adult (BR-005); a record of every attempt; and progress summaries for supporting adults. Supporting adults are parents, guardians, caregivers, educators and learning-support staff. They hold the accounts, choose lessons and starting stages, and decide when a child moves on (BR-001, BR-003, CON-007). Squigglr gives no diagnosis, no judgement of a child's abilities and no comparison with other children (BR-006).

Several conditions of use shape Squigglr's hazards. Children may practise without an adult watching (ASM-002), may not be able to read (FACT-007), and may share one device with other children (ASM-008). Adults decide on a child's practice from the records (ASM-006), so a record that misstates what the child did can mislead them (BR-004). Creative drawings may be refined with an existing pretrained model that the team does not train or run (Development Plan, SRC-002; LC-004). The website also depends on hosting, sign-in and network services outside the team's control and, at school, on information technology (IT) staff who manage the devices and networks (PER-3, ASM-013).

## 1.2 Purpose and Scope of this Document

This document records the hazards the team has identified, the failure modes behind them, and the action recommended for each. Its outputs are safety requirements: every recommended action traces to a requirement in the SRS, and the requirements derived from this analysis are listed again in Section 6.

The team uses the analysis to decide what the SRS must require and what later verification must test; reviewers use it to check that the requirements address the ways Squigglr can fail. The analysis covers Squigglr as the SRS specifies it, used at home, at school and in other places where children practise. Section 2 states the losses considered and what lies outside the scope, and Section 3 draws the system boundary.

## 1.3 Definition of Hazard

This analysis uses the definition of hazard from Nancy Leveson's work that the course adopts (Smith, 2026). In the wording of Leveson and Thomas (2018, p. 17), a hazard is "a system state or set of conditions that, together with a particular set of worst-case environmental conditions, will lead to a loss". A loss "involves something of value to stakeholders" (Leveson and Thomas, 2018, p. 16). For Squigglr, the system is the set of components inside the boundary of Section 3, and the environment is everything outside it: the child, the supporting adult, the device, the network and the outside services. For example, a learner profile left active after one child's turn on a shared school tablet is a hazard. When the next child draws without an adult noticing (the environmental condition), that child's attempts are saved to the wrong record, and an adult later reads a history that mixes two children (a loss of privacy and of correct records).

Squigglr controls no physical equipment, so the course's "weakened" notion of harm applies: besides injury, harm includes security failures, insensitivity to users, frustrating software and poor data quality (Smith, 2026). Section 2 turns this into five losses specific to Squigglr. Safety is also not the same as correctness (Smith, 2026): a component can meet its specification and still contribute to a loss in a harmful situation, so the analysis looks at conditions of use as well as defects.

## 1.4 Changes to the Template

The course template has been changed as follows. Every template section is kept, in its original order.

- *Front matter:* a List of Figures and a List of Tables follow the Table of Contents, and the template's guidance notes are removed.
- *Introduction:* divided into background, purpose and scope, the definition of hazard, these changes and a document roadmap.
- *Scope and purpose (Section 2):* the losses are listed with IDs L1 to L5, and each item left out of scope is given a reason and a stance on its hazards.
- *System boundaries and components (Section 3):* the components have IDs C1 to C8 and fixed names, which the Component column of the H-MUH table and the coverage table of the Failure Mode and Effect Analysis

(FMEA) repeat exactly. A figure shows the boundary, and a table states how failures outside it enter the analysis. C8 is added to the seven components first planned, for the reason given in Section 3.

- *FMEA (Section 5):* each table has the columns Hazard ID, Component, Failure mode, Cause(s), Effect and loss, Category, Recommended action and Mitigating requirement IDs. The Category column is added; it classes each row as Physical, Psychological/well-being, Privacy/security, Data integrity or Availability. Each team member's rows form a subsection with a table of their own, so each hazard ID carries its writer's prefix: Table 5 (H-RAY), Table 6 (H-AHM), Table 8 (H-MOU) and the H-MUH table. True duplicates are combined by reference to one row. A coverage table then checks every component of Table 3 against every category, across all the tables. Each recommended action begins with Eliminate, Mitigate or Document/Test. Rows are not ranked or filtered by a risk priority number, following the course's advice against prioritising hazards by probability (Smith, 2026).
- *Safety and security requirements (Section 6):* as the teaching assistant (TA) advised (SRC-006), each safety requirement is written once, in the SRS, and the FMEA refers to it by ID. This section repeats the statement of every requirement derived from the HA exactly as the SRS states it, with the hazards it mitigates.
- *Roadmap (Section 7):* each requirement derived from the HA is assigned to a project phase, PH-1 (proof of concept), PH-2 (Revision 0) or PH-3 (final), or to Future, with the phase dates set in the SRS Introduction.

### 1.5 Document Roadmap

Section 2 states the purpose of the analysis, the losses it considers and what is out of scope. Section 3 divides Squigglr into components and draws the boundary between them and the people, equipment and services around them. Section 4 lists the critical assumptions, kept few so that they do not assume hazards away. Section 5 states the method, gives the FMEA tables, one for each author's rows, and checks their coverage. Section 6 lists the safety and security requirements derived from the analysis, and Section 7 gives the phase in which each is planned. The references follow, then two appendices: the disclosure of generative artificial intelligence (AI) use and the team's reflection. Table 1 records the changes made to this document.

## 2 Scope and Purpose of Hazard Analysis

The purpose of this analysis is to find the ways Squigglr's components can fail, or can work as specified yet still contribute to harm in some situation, and to recommend for each an action that eliminates the hazard, mitigates it, or

documents and tests it (Smith, 2026). The actions become requirements in the SRS, so the analysis is part of specifying Squigglr, not a check made after it is built. The analysis covers Squigglr as the SRS specifies it for goals G.1 to G.6, used by children and supporting adults wherever children practise: at home, in classrooms, and in places such as rehabilitation centres (SRS, Anticipated Workplace Environment). Hazards are identified per component, so the scope of the analysis is the system boundary of Section 3.

## 2.1 Losses Considered

Table 2 lists the five losses this analysis considers. For each, it states what the loss means for Squigglr, gives an example and cites the evidence that the loss can occur. Every example uses invented, not real, children and adults.

Table 2: Losses considered in the hazard analysis

| ID | Loss | Meaning for Squigglr | Example | Evidence |
|---|---|---|---|---|
| L1 | Physical harm | Bodily harm to a child that Squigglr's own behaviour contributes to. Squigglr controls no equipment, and an adult nearby stays responsible for the child's physical safety and time on the device (ASM-002). The analysis therefore considers only what Squigglr does: content on a child-facing screen that flashes and can trigger a seizure in rare cases (H-MOU-001), and practice that does not end or restarts at once and keeps a child on the device far longer than the adult intended. | Feedback on a child-facing screen flashes several times a second in front of a child who is prone to seizures; or a child practising alone restarts a lesson each time it ends at its attempt limit and keeps drawing for much longer than planned. | ASM-002; G.3; WCAG 2.2 criterion 2.3.1 (World Wide Web Consortium, 2023); HA checklist (addictiveness) |
| L2 | Psychological harm and well-being | Distress, frustration, agitation, or loss of confidence or of ownership of the work, caused by what Squigglr shows or does. | A child who is agitated by repeated tasks fails the same stage again and again; or a refined drawing no longer feels like the child's own and the original cannot be restored. | PER-1; BR-005; ASM-011; ASM-015; HA checklist (well-being, self-image) |
| L3 | Privacy loss | A child's practice records or drawings reach a person or service that should not have them. Records describe a child's difficulties and are personal information (BR-001). | On a shared school tablet, one learner's records are shown to another learner; or a drawing sent for refinement is kept or reused by the outside service. | BR-001; ASM-008; ASM-009; PER-3; OI-02; SRC-002 |

| ID | Loss | Meaning for Squigglr | Example | Evidence |
|---|---|---|---|---|
| L4 | Data loss or corruption | An attempt, lesson record, creative drawing or setting is lost, altered, saved to the wrong place, or no longer matches what happened. | The connection drops as a child finishes an attempt and the attempt is never saved; or refinement overwrites the original strokes so that they cannot be restored. | ASM-005; G.4; G.5; BR-004; BR-005 |
| L5 | Misleading information leading to poor support decisions | What Squigglr shows an adult leads them to a wrong decision about the child's practice or support. | A summary counts attempts made with a tracing guide as made without one, so the adult moves the child on too soon; or an adult reads the records as an assessment of the child. | ASM-006; G.6; BR-004; BR-006; H-RAY-003 |

The Category column of the FMEA (Section 5) classes each row as Physical, Psychological/well-being, Privacy/security, Data integrity or Availability. The first four categories correspond to losses L1 to L4, and a Data integrity or Psychological/well-being row also leads to L5 when its effect is a misleading record or summary. Availability has no loss of its own: when Squigglr or a service it depends on cannot be used, the losses are lost practice (L4) and frustration (L2). Each row's Effect and loss column names the loss it leads to.

## 2.2 Out of Scope

The items below are outside the scope of this analysis. Losses or items that are excluded should be documented (Leveson and Thomas, 2018, p. 17), so each item has a reason and a stance on what happens to its hazards.

- **Camera-based finger tracing.** A stretch goal (SRC-001). The specialists stated that tracing in the air practises gross rather than fine motor skills and should come after the touchscreen activities (FACT-006), and no core requirement uses a camera. *Stance:* a camera would capture images of the child, while a learner profile holds only a first name and a non-identifying username (BR-001). Before camera tracing is added, this analysis must gain a camera component and analyse its privacy hazards; nothing here covers them.
- **Letter tracing and drawing.** A stretch goal (SRC-001); two specialists saw letters as a later stage after shapes (ASM-010; SRC-009.04). *Stance:* letters would be new lessons in the same catalogue (LC-001), so the hazards of the canvas, lesson and storage components apply to them unchanged. Letter-specific hazards, such as an adult reading letter attempts as a handwriting assessment (BR-006), must be added when letters are specified.

- **Finger input as a supported input.** A stretch goal (CON-003, LC-006); stage checks and value ranges are set for stylus input (ASM-003). *Stance:* drawing with a finger on a device without a stylus remains possible, and is in scope as a condition of the device outside the boundary, since such attempts may be judged wrongly (ASM-003). Only finger input as a supported feature, with its own ranges, is out of scope; adding it requires the canvas and lesson rows to be reviewed.

- **Offline use.** Out of the initial scope (ASM-005). *Stance:* a slow or lost connection is in scope as a condition of the network outside the boundary. Only an offline mode, in which Squigglr works without a connection, is out of scope.

- **Internal operation of outside equipment and services.** The device, browser, network, hosting service, authentication service and refinement model are not analysed internally, because the team neither builds nor controls them. Physical harm from the device itself, such as a dropped tablet, stays with the adult nearby (ASM-002). *Stance:* their failures are in scope at the boundary, as causes of failure modes of the components that use them (Section 3).

- **The project's own activities.** Requirements gathering and evaluation involve no children (CON-005, CON-006). These are constraints on the project, not behaviour of Squigglr. *Stance:* any future study with children needs ethics approval first (FACT-008), which this analysis does not replace.

Diagnosis and treatment are never features of Squigglr (BR-006), so no hazard of a diagnostic function is analysed. The use of practice records as diagnostic evidence, however, is in scope, because the records exist and adults may show them to others (loss L5; H-RAY-003).

# 3 System Boundaries and Components

The system analysed is Squigglr: the functions the team specifies and builds, wherever they run. FMEA is applied to functional rather than physical components (Smith, 2026), so the components below are parts of Squigglr's behaviour, not modules of a design; none names a framework, a data schema or an interface. Figure 1 shows the eight components inside the boundary and the people, equipment and services outside it. Table 3 describes each component, and Table 4 describes each outside item and how its failures enter the analysis.

## 3.1 Components Inside the Boundary

Table 3 gives each component's ID and name, its role, what it holds or decides, and the SRS items it relates to. The names are fixed: the Component column of the H-MUH table of the FMEA and its coverage table repeat them exactly;

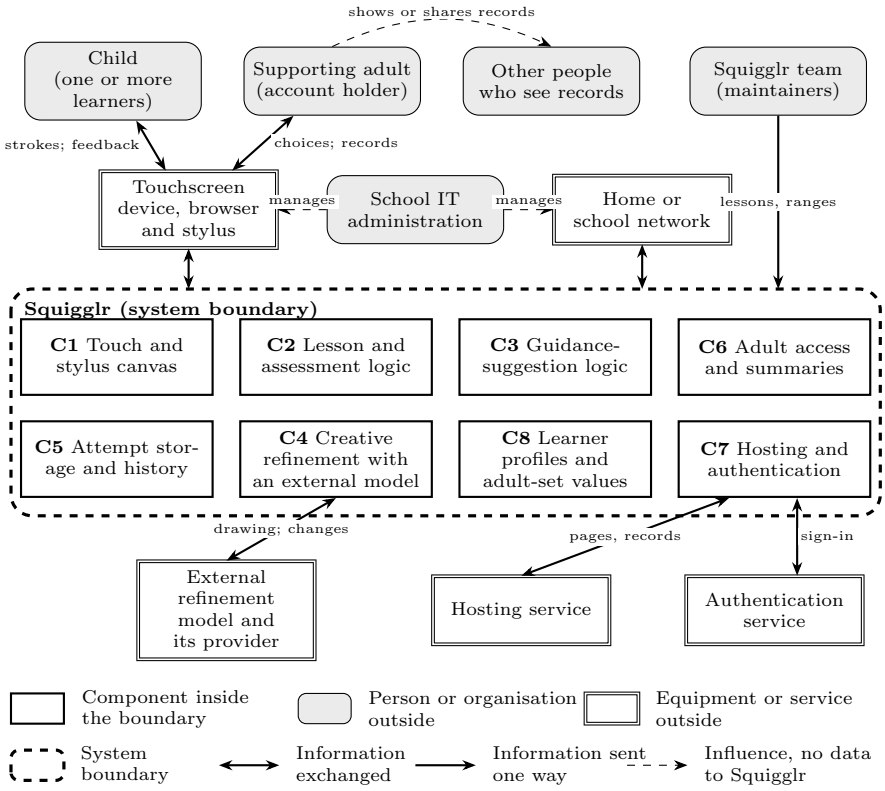



Figure 1: System boundary of Squigglr: the components inside it and the people, equipment and services outside it. Arrow labels name the main information exchanged.

the H-RAY and H-AHM tables name functions in their authors' words, and the coverage table assigns each of their rows to one of these components. The components follow the order in which a child's work passes through Squigglr, from drawing to records and adult review, with C8 last because it was added later.

Table 3: Components inside the system boundary

| ID | Component | Role | Holds or decides | Related SRS items |
|---|---|---|---|---|
| C1 | Touch and stylus canvas | Takes the strokes a child draws, shows them as drawn, and presents each stage's demonstration, tracing guide and feedback, on screen and as sound. In the creative space, carries out the child's edits (move, resize, recolour, erase) and their undo. | The strokes of the current attempt or drawing until they are checked or saved. | G.1, G.4; CON-003; ASM-003, ASM-004; UC-003, UC-006 |
| C2 | Lesson and assessment logic | Runs each lesson's stages, checks every attempt against the stage's pass criteria, chooses predefined feedback, moves the child on after a pass, and ends the lesson at the attempt limit, extending it once if the child is improving (BR-002). | Whether an attempt passes; the predefined difficulties observed; when a lesson ends. | G.1, G.2, G.3; BR-002, BR-004, BR-007; UC-003 |
| C3 | Guidance-suggestion logic | Decides when to suggest going back a stage after repeated failed attempts, unless the child is improving, holds the suggestion until an adult responds, and applies it only if the adult accepts. | Each suggestion's state: pending, accepted, declined or ignored. | G.3; BR-003; CON-007; UC-005 |
| C4 | Creative refinement with an external model | Refines a creative drawing minimally using an external refinement model, keeps the result editable, and restores the child's original strokes when the child asks or, if the child cannot, when an adult asks. Covers what Squigglr sends to the model and how it checks what comes back. Refinement can be turned off (UC-007). | Each creative drawing's original strokes and refined version. | G.4; BR-005; ASM-011; LC-004; UC-006, UC-007 |
| C5 | Attempt storage and history | Saves each attempt's record (stage, strokes, predefined difficulties, corrections, outcome) and each lesson record, keeps creative drawings, files every record under the right attempt and learner, and returns the history when asked. | Every saved record and its link to an attempt and a learner. | G.5; BR-002, BR-004; ASM-005; UC-008 |

| ID | Component | Role | Holds or decides | Related SRS items |
|---|---|---|---|---|
| C6 | Adult access and summaries | Gives supporting adults the views and actions only they may use, behind an adult check that a child cannot complete by tapping icons (SAF-RAY-001): choosing a lesson and starting stage, switching the active learner, responding to suggestions, changing adult-set values, restoring an original drawing for a child, and reading progress summaries and the attempts behind them. | Which actions need an adult; what each summary shows. | G.3, G.6; BR-001, BR-003, BR-006; ASM-006; UC-002, UC-004, UC-005, UC-009, UC-010 |
| C7 | Hosting and authentication | Squigglr's use of a hosting service to serve the website and keep its records, and of an authentication service to sign in supporting adults and keep each account's data to that account. | Which account is signed in; which account owns each learner profile and record. | BR-001; CON-001, CON-002; ASM-005, ASM-009, ASM-013 |
| C8 | Learner profiles and adult-set values | Holds the learner profiles under an account, keeps track of which learner is active, and stores each child's adult-set values, accepting only values within their ranges, and other settings such as muted audio (ASM-015) and whether refinement is on (UC-010). | Each profile's first name and non-identifying username; the active learner; the values in use. | BR-001, BR-008, BR-009; ASM-008, ASM-016; UC-001, UC-010 |

Two components need a note. C4 and C7 cover Squigglr's *use* of outside services: what it sends, what it accepts back and how it behaves when a service fails. The services themselves lie outside the boundary. C8 was added to the seven components first planned because learner profiles and adult-set values follow rules of their own (BR-001, BR-008) and are used by several components: C2 applies the values, C5 files records under the active learner, and C6 lets an adult change them. Failures in choosing the active learner on a shared device, or in storing and checking an adult-set value, would otherwise be split among those components; C8 gives them one component in the FMEA coverage table.

## 3.2 Outside the Boundary

The items in Table 4 lie outside the boundary because the team neither builds nor controls them. Leaving an item outside does not mean it is assumed to work. Its failures enter the FMEA as causes of failure modes of the components that meet it, named in the last column of the table, and Section 4 states the few assumptions made about these items.

Table 4: Items outside the system boundary and how their failures enter the analysis

| **Outside item** | **Why it is outside** | **How its failures enter (components)** |
|---|---|---|
| Child (one or more learners) | Uses Squigglr but is not controlled by it: what the child draws, how they react, and whether the right learner is active are outside Squigglr (ASM-002, ASM-008). | Unexpected or partial input (C1); an adult-only action reached by a child (C6); drawing under another learner's profile (C8); distress at feedback or refinement (C2, C4). |
| Supporting adult | Chooses lessons and values and reads the records; their judgement and attention are outside Squigglr (ASM-006, ASM-016). | Unsuitable values (C8); a response to a suggestion (C3); a misread summary (C6); not watching the child (C2, C6). |
| Other people who see records | People who do not hold the account, such as other educators, therapists, family members or other learners at a shared device. Squigglr has no relation with them (BR-001). | Records read as an assessment of the child (C6); records shown to someone who should not see them (C6, C8). |
| Touchscreen device, browser and stylus | Owned and chosen by families and schools; the supported combinations are not yet defined (CON-003, OI-09, ASM-003). | Missing or coarse stroke data (C1; ASM-004); an unsupported device or no stylus (C1); a session cut off when the browser closes or the device stops (C1, C5). |
| Home or school network | Outside the team's control; Squigglr needs a connection to load lessons and to save and view records (ASM-005). | A slow or lost connection during an attempt, a save or a refinement (C5, C4, C7). |
| School IT administration | Manages school devices and networks, on which adults may not be able to install software (CON-002); a school may block the site (ASM-013). | Squigglr unavailable or partly blocked at school (C7, C1). |
| External refinement model and its provider | An existing pretrained model that the team does not train or run, from a provider that may change (SRC-002; LC-004). | The service unavailable or slow, changes that are invalid or too large, or drawings sent to it kept by the provider (C4). |
| Hosting service | A managed service that serves the website and stores the records (SRC-002). | An outage, or stored records lost or delayed (C7, C5). |
| Authentication service | A managed service that signs in supporting adults (SRC-002). | Sign-in unavailable, or an adult's account taken over (C7, C6). |
| Squigglr team (maintainers) | Supplies the lesson catalogue, the ranges of the adult-set values and the fixed values (BR-009) outside any use of Squigglr. | A wrong lesson, range or fixed value (C2, C8). |

The SRS describes the context of the work, in its Scope of the Work section, through the actors that exchange business information with it: the child, the supporting adult and the Squigglr team. This analysis also shows the equipment,

network and outside services, because they carry or process the child's data and their failures can cause the losses of Table 2.

# 4 Critical Assumptions

The assumptions below are kept few and are written so that they do not remove hazards. Squigglr does not assume that an adult watches the child during practice, that the network is always available, or that the device meets Squigglr's requirements; those conditions are treated as failure modes in the Failure Mode and Effect Analysis (FMEA). Sources are cited by their identifiers in the Software Requirements Specification (SRS), Appendix A, and interview turns by speaker and timestamp, for example (SRC-005, P1, 34:13). Each assumption lists the H-RAY hazards it affects.

**CA-001** An adult sets up Squigglr for a child, chooses and starts the lesson and its starting stage, and is the only person meant to accept a suggestion to go back a stage. Within a lesson the child does not navigate, exit or restart; Squigglr moves between stages. An adult may not be present or watching while the child practises.
*Rationale:* children should practise without constant adult help, while adults handle setup (SRC-005, P1, 34:13), and adults decide progression (SRC-005, P1, 28:37; CON-007).
*Hazards affected:* H-RAY-001, H-RAY-004, H-RAY-005.

**CA-002** The device, browser, touchscreen or stylus in use may not be one of the supported combinations (`SUPPORTED_DEVICES`, TBD), and Squigglr may be opened on a device without a stylus, where the child can draw with a finger but not as intended (ASM-003).
*Rationale:* Squigglr is a website opened from a web address with no installation (CON-001, CON-002), so the team does not control the device. The supported combinations are not yet defined (CON-003), and the team did not promise that Squigglr works on every device (SRC-005, Moustafa, 6:03).
*Hazards affected:* H-RAY-002.

**CA-003** The network connection may be slow or lost at any time, including during an attempt.
*Rationale:* attempts are saved to an account and reviewed later from another device, and offline access was not promised (SRC-005, Moustafa, 6:03). Home and school networks are outside the team's control.
*Hazards affected:* H-RAY-002, H-RAY-006.

**CA-004** One device may be used by several children, at school or at home.
*Rationale:* school devices may be shared by several children (SRC-004), and Squigglr may be used at home, at school or in other settings such as rehabilitation centres (SRC-005, P2, 21:25).
*Hazards affected:* H-RAY-006.

**CA-005** The adults who read practice records are not assumed to have training in visual-motor development, and they may show the records to other people, such as teachers or therapists.
*Rationale:* parents and caregivers are not assumed to be specialists (SRS, Hands-On Users), and records exist to support conversations about what a child needs (SRC-005, Moustafa, 6:03). Squigglr makes no diagnosis or treatment claims (SRC-001).
*Hazards affected:* H-RAY-003.

**CA-006** The supporting adult chooses the adult-set values, such as `STAGE_PASS`, `DIFFICULTY_STREAK` and `ATTEMPT_LIMIT`, for each child within a range the team sets (SRC-008). The ranges are not found from child testing, and the value an adult picks, or the default, may not suit their child.
*Rationale:* no information is gathered from children and no study with children is run (CON-005, CON-006), so the ranges cannot be tuned on children (SRS, Goals of the Project). Children reach these skills at different points (SRC-005, P1, 20:27).
*Hazards affected:* H-RAY-004, H-RAY-005, H-RAY-007.

# 5 Failure Mode and Effect Analysis

This section applies Failure Mode and Effect Analysis (FMEA) to the components of Squigglr. Subsection 5.1 states how the analysis was carried out. Each team member's rows are in a table of their own: Table 5 (H-RAY), Table 6 (H-AHM), Table 8 (H-MOU) and Table 9 (H-MUH). Table 10 places every row of the four tables under a component and checks that every component has been examined for every category of harm. Each row ends with the requirements of the Software Requirements Specification (SRS) that mitigate it; the Safety and Security Requirements section lists the requirements derived from this analysis again.

## 5.1 Method

FMEA is a bottom-up method: it starts from the functions of each component and asks how each can fail (Smith, 2026). The components are the functional components named in the System Boundaries and Components section, not modules of a design. The analysis uses the course's definitions (Smith, 2026):

- a *failure* is a departure of a component from its required or intended function;
- a *failure mode* is the manner in which the failure is observed by a child, an adult or a tester;
- an *effect* is the consequence of a failure mode; each row names the losses it leads to, and the H-MUH rows use the IDs L1 to L5 of the Scope and Purpose section;

- a *cause* is a reason the failure mode occurs, including conditions outside the boundary, such as a lost connection (CA-003) or a device shared by several children (CA-004);
- a *recommended action* says what is done to eliminate or mitigate the hazard, and becomes a safety requirement.

Failure modes were found in three steps. First, the functions of each component were taken from its role in the System Boundaries and Components section, and four guide words were applied to each function: it is not performed; it is performed wrongly or only in part; it is performed at the wrong time or out of order; and it is performed when it should not be. Second, every component was checked against the five categories of harm (Physical, Psychological/well-being, Privacy/security, Data integrity and Availability), so that a component that works as specified but still harms a child is also recorded; safety is not the same as correctness (Smith, 2026). For the creative refinement component, which uses an external model, this check also covered the risks of artificial intelligence (AI) that arise from how Squigglr uses the model: compromise of privacy, over-reliance, loss of human agency, lack of capability or robustness, and lack of transparency (Slattery et al., 2024). Third, each team member writes the rows for the parts of the SRS they own, so each hazard ID carries its writer's prefix (H-RAY, H-MOU, H-AHM, H-MUH or H-SHA), and each member's rows form a table of their own. The coverage check of Subsection 5.7 then places every row of every table under one of the components, so that the rows of all members are checked against the same components and categories.

Every row has a recommended action. Each action begins with *Eliminate* (remove the failure mode, for example by not offering a function where it can cause the harm), *Mitigate* (reduce its effect, or make it visible and recoverable) or *Document/Test* (a statement to users or a verification activity that limits or checks what remains); one cell may hold several. Actions state what Squigglr must do, not how. The last column of each row names, by ID, the SRS requirements, goals or constraints that carry out its action. For the H-MUH rows, each cited requirement lists the hazard among its own traces, so the trace runs in both directions. An action that needs a requirement not yet in the SRS is added to the owner's section of the SRS, or to its Safety-Critical Requirements subsection, before its ID appears in the table.

Rows are not ranked, and no row is left out because its failure seems unlikely. Many FMEAs rank rows by a risk priority number, the product of ratings for severity, probability of occurrence and detection, and mitigate only the rows above a threshold. The course advises against this, because accidents have followed from judging that a failure is not likely to happen, and there is seldom a good basis for the probabilities (Smith, 2026). Squigglr has no usage data from children on which to base them (CON-005, CON-006), so every row is mitigated.

Rows from different owners can describe the same failure. When two rows share a failure mode and a cause, they are combined by reference: one row keeps the shared causes and their mitigations, and the other states that it is combined

with that row, keeps only the causes that are its own, and keeps its ID, so that no requirement's trace is broken. In Table 9, rows are grouped by component in the order of the components table of the System Boundaries and Components section, and by ID within a group. Where a row's causes also lie in another component, the coverage check of Table 10 records it under that component too.

## 5.2 Supervision, Context of Use and Use of Records (H-RAY)

Table 5 covers hazards from how Squigglr is supervised, where and on what it is used, and how its records are read. Each recommended action starts with Eliminate, Mitigate or Document/Test. Mitigating requirements trace to the goals (G.#) and constraints (CON-###) in the SRS, or to safety requirements SAF-RAY-### in the SRS, Safety-Critical Requirements.

Table 5: FMEA: supervision, context of use and use of records

| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| H-RAY-001 | Lesson flow; adult controls | A child practises with no adult watching, or the adult is inattentive, and the child reaches an adult-only action. | Children are meant to practise without constant adult help (CA-001); adult controls use the same touch interface as the child. | The child accepts a suggestion to go back a stage, or changes lesson or stage, without an adult's decision; the child's distress goes unnoticed. Loss: progression not chosen by an adult; well-being. | Psychological/ well-being | Mitigate: place every adult-only action (choosing a lesson or stage, accepting a suggestion) behind an adult check that a child cannot pass by tapping icons; keep a pending suggestion waiting, without blocking practice, until an adult reviews it. | CON-007, G.3, SAF-RAY-001 |
| H-RAY-002 | Browser front end; saving of attempts | Squigglr is used on an unsupported device or browser, without a stylus, or loses its connection during an attempt, at home, at school or on a shared device. | Device not controlled by the team (CA-002); no offline access (CA-003). | Strokes are not captured correctly, the child's attempt is lost, or a lesson cannot be completed; the child is frustrated and the record is incomplete. Loss: lost practice data; availability. | Data integrity or Availability | Mitigate: before a lesson starts, check the device and browser against `SUPPORTED_DEVICES` and show the adult a notice when they do not match; keep each attempt until the server confirms it is saved, and tell the adult when an attempt could not be saved. | CON-001, CON-002, CON-003, G.5, SAF-RAY-002, SAF-RAY-003 |

| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| H-RAY-003 | Attempt records; progress summaries | Practice records are used as clinical or diagnostic evidence, or as a measure of the child's ability. | Adults reading records lack specialist training, and records may be passed to others (CA-005); fewer attempts may be read as better skill (SRC-005, Moustafa, 6:03). | An adult or professional draws a wrong conclusion about the child, such as a label or a placement decision. Loss: harm to the child's well-being and to decisions about support. | Psychological/ well-being | Eliminate: show no diagnostic labels, judgements of the child's abilities or comparisons with other children in any record or summary. Mitigate: state on every summary and export that the records describe practice only and are not an assessment. | G.5, G.6, SAF-RAY-004 |
| H-RAY-004 | Lesson flow; feedback | Repeated failed attempts at one stage continue until the child becomes frustrated or agitated. | Failure feedback repeats; `ATTEMPT_LIMIT` may be too high for a given child (CA-006); an improvement extension lengthens the lesson; no adult may be watching (CA-001). | The child becomes distressed, stops practising or refuses the activity later. Loss: well-being; the practice benefit. | Psychological/ well-being | Mitigate: end the lesson at `ATTEMPT_LIMIT`, or at `ATTEMPT_LIMIT_MAX` when extended, and save progress (G.3); let audio feedback be muted (SRC-005, P2, 37:39). Document/Test: test that each lesson ends at `ATTEMPT_LIMIT` and that no extended lesson passes `ATTEMPT_LIMIT_MAX`. | G.3, SAF-RAY-005 |

| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| H-RAY-005 | Lesson flow; attempt records | After a lesson ends at `ATTEMPT_LIMIT`, it is restarted at once and the repetition continues. | No adult may be watching (CA-001); a restart starts a new count of attempts. | The limit does not stop the repetition, and the adult cannot see how often it happened. Loss: well-being; misleading records. | Psychological/ well-being | Mitigate: record each lesson that ended at `ATTEMPT_LIMIT` and show the adult how many times this happened per lesson; keep earlier attempts when a lesson is restarted. Document/Test: test that a restart after the limit keeps the earlier record. | G.3, G.5, SAF-RAY-006 |
| H-RAY-006 | Learner selection; attempt records | On a shared device, attempts or a pending suggestion are saved to, or shown for, the wrong child. | Several children use one device (CA-004); the active child is not changed between sessions; a lost connection leaves a session open (CA-003). | One child's records are shown to another child or adult, or a child's history is mixed with another's, so an adult accepts a suggestion based on the wrong attempts. Loss: privacy; record integrity. | Privacy/ security | Mitigate: show which learner is active on every practice screen in a way a child who cannot read can recognize; require an adult action to switch learners; show the attempts behind a suggestion when the adult reviews it. | G.5, SAF-RAY-001, SAF-RAY-007 |

| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| H-RAY-007 | Child profile settings; lesson flow | A stage check or lesson limit uses a value that does not suit the child: `STAGE_PASS` too strict or too lenient, or `ATTEMPT_LIMIT` not larger than `DIFFICULTY_STREAK`, or `ATTEMPT_LIMIT_MAX` below `ATTEMPT_LIMIT`. | The adult misjudges the example drawings or leaves a default that does not suit the child (CA-006); a value is saved outside its range; a child changes a setting. | A child cannot pass a stage they can draw, or passes one they cannot, and the records mislead the adult. Loss: well-being; misleading records. | Psychological/ well-being | Mitigate: accept only values within the team-set range, and only an `ATTEMPT_LIMIT` larger than `DIFFICULTY_STREAK`; show example drawings of each end of the `STAGE_PASS` range; place settings behind the adult check; show the values in use next to the child's records. Document/Test: test that out-of-range and invalid values are refused and that the default is the midpoint. | G.2, G.3, SAF-RAY-008 |

### 5.3 Drawing Input and Guided-Lesson Decisions (H-AHM)

Table 6 examines failures in the capture and interpretation of stylus input and in the decisions that use it. These failures can occur even on an otherwise supported device with a working network, so they are distinct from the environmental failure in H-RAY-002. Effects are assessed against drawing accuracy, record integrity and the child's well-being; Squigglr does not make clinical assessments (BR-006). Recommended actions are traced to the learning-activity requirements in the SRS. Two additional safeguards, SAF-AHM-001 and SAF-AHM-002, are specified in the SRS Safety-Critical Requirements section.

Table 6: FMEA: drawing input and guided-lesson decisions

| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| H-AHM-001 | Stylus input capture; attempt state | A stroke or part of one stroke disappears during a multi-stroke attempt, but the attempt is still finalized and evaluated. | Missed pointer samples, a brief input interruption or premature finalization (ASM-004); a stylus lift is mistaken for the end of an attempt. | The drawing appears incomplete or is scored inaccurately; a child receives undeserved failure feedback or loses a practice attempt. Loss: accurate feedback and record integrity. | Data integrity | Mitigate: retain ordered points and stroke boundaries; distinguish stylus lifts from finalization; treat interrupted or unevaluable input as non-passing without consuming an attempt. Document/Test: replay partial strokes and interruptions before and during a stroke. | FR-LRN-007, FR-LRN-008, FR-LRN-009, FR-LRN-010 |
| H-AHM-002 | Stylus event capture; attempt evaluation | One drawn stroke is captured or evaluated twice, inflating coverage or producing a spurious extra line. | Duplicate input events, accidental replay after reconnection, or combining the same stroke twice. | A poor attempt may pass, or a correct one may fail; saved strokes no longer represent what was drawn. Loss: integrity of feedback and practice records. | Data integrity | Mitigate: reject duplicate events within the same attempt before evaluation, without deleting legitimate separate strokes. Document/Test: replay duplicate events and distinct repeated strokes; verify that each is handled appropriately. | FR-LRN-007, FR-LRN-008; SAF-AHM-001 |

| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| H-AHM-003 | Input coordinates; reference-shape alignment | Valid stylus points are mapped to the wrong position or order and used to score the drawing. | Wrong canvas scale, coordinate offset or incorrect ordering after a resize or orientation change (ASM-004). | An accurate attempt fails, an inaccurate attempt passes, or incorrect strokes are saved. Loss: data integrity and trustworthy progress records. | Data integrity | Mitigate: verify that captured coordinates remain in the current drawing surface and retain their recorded order; reject input that cannot be mapped reliably. Document/Test: repeat known stroke traces before and after resizing or rotating the device. | FR-LRN-007, FR-LRN-010; SAF-AHM-002 |
| H-AHM-004 | Completeness and pass classifier; lesson flow | An incomplete drawing (for example, a square missing a side) is reported as passing and advances the child. | The geometric result is high even though a required segment is missing, or completeness is not checked before threshold comparison. | The system advances despite an unfinished task and records a misleading success. Loss: record integrity and appropriate guidance. | Data integrity | Eliminate: require completeness and the adult-selected passing threshold to both hold before a pass; otherwise remain at the same stage. Document/Test: use complete and missing-part drawings immediately above and below the threshold, including a high-scoring incomplete example. | FR-LRN-011, FR-LRN-013, FR-LRN-023 |

| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| H-AHM-005 | Feedback-selection logic; stage-result mapping | A detected attempt result or drawing difficulty is associated with the wrong pre-approved feedback observation (for example, a missed endpoint is reported as a missing segment). | A classification identifier is mapped to the wrong feedback entry, or the feedback lookup uses the wrong lesson or guidance stage. | The child is directed to correct a part of the drawing that was not identified as incorrect; the reported observation does not match the evaluated attempt. Loss: accurate guidance and feedback integrity. | Data integrity | Mitigate: select feedback from the approved catalogue using the evaluated result and correct lesson and stage identifiers; when no matching observation exists, do not present an unrelated correction. Document/Test: check the mapped message for every predefined classification and each drawing stage; test missing and mismatched identifiers. | FR-LRN-011, FR-LRN-013, FR-LRN-014 |
| H-AHM-006 | Failure-streak and suggestion logic; adult decision | Squigglr recommends moving backward at the wrong time, or changes the stage without the adult accepting the suggestion. | Invalid input is counted as failure, the streak is not reset, or the improvement suppression rule is omitted (BR-003). | The child receives unsuitable support or a needless interruption; adult control is bypassed. Loss: well-being and authorized lesson progression. | Psychological/ well-being | Mitigate: base the suggestion only on counted non-passing attempts and the defined improvement predicate; leave the current stage unchanged until an authorized adult accepts. Document/Test: replay improving, non-improving, invalid and threshold-boundary sequences; decline or ignore a suggestion. | FR-LRN-009, FR-LRN-018, FR-LRN-019, FR-LRN-020 |

| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| H-AHM-007 | Learner settings; attempt and lesson evaluation | The active attempt is evaluated using another learner's threshold, a stale value or a setting different from the adult's saved selection. | Wrong profile linkage, settings not refreshed after an adult change, or defaults applied over saved choices (BR-008; CA-006). | A drawing passes or fails incorrectly, a lesson ends too early or continues too long, and the adult receives misleading records. Loss: well-being and record integrity. | Data integrity | Mitigate: use the active learner's validated saved settings for passing, suggestions and limits; use midpoint defaults only for unset values. Document/Test: switch learner profiles and change settings; test boundaries, invalid combinations and the exact saved-versus-default choice. | FR-LRN-016, FR-LRN-017, FR-LRN-019, FR-LRN-021 |

### 5.4 Security, Saving and AI-Assisted Refinement (H-SHA)

Table 7 covers hazards regarding account and data security, practice records, and the external AI refinement provider. Effects are assessed based on privacy, data integrity, and the trustworthiness of content shown to children and adults.

Table 7: FMEA: security and AI-assisted refinement

| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| H-SHA-001 | Authentication; account management | An unauthorized user gains access to a learners account data or the entire account. | Weak or reused credentials; missing server-side authorization checks | The child's privacy is violated as the unauthorized user can read or write over personal data. Loss: privacy and trust in the system. | Privacy/security | Mitigate: implement authentication at every endpoint the data is accessed at; encrypt user data with one-way hash; end session and logout after a extended inactivity; Document/Test: test that attempts to access other accounts' data under the hood is rejected by the server | SAF-SHA-001 |
| H-SHA-002 | Data transmission; third-party integrations | Private learner data is sent to an third-party service (other than ones approved for functionality) | Error reporting or refinement service receive raw request payloads of private learner data; developers use unininspected external SDK | Privacy policy violated and child's private data exposed. Loss: privacy; regulatory compliance. | Privacy/security | Mitigate: administer protocol for reviewing all third-party services that are essential before implementing, reject all non-essential services. Document/Test: run grey-box (white-box if open source) integration tests in contained environments inspecting whether third-party services transmit PII. | SAF-SHA-002 |

| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| H-SHA-003 | Attempt persistence; save confirmation | A practice attempt is lost without the user being informed, or the system reports a false successful save. | Network error after rendering of success response; network error without notification | Adult believes child's progress is being tracked with complete records when records are missing in reality. Loss: misleading progress reporting; adult trust in the application. | Data integrity | Mitigate: confirm a save with a silent verification ping that informs the adult if the save did not really save; Document/Test: run integration testing wherein saves are accompanied by simulated errors and verify that it is reported. | G.3, SAF-SHA-003 |
| H-SHA-004 | AI refinement provider; lesson content delivery | The external AI refinement service is unavailable, returns an error, or returns unintelligible content. | Provider outage; rate-limiting; response schema mismatch due to update. | The child's lesson cannot proceed or is shown incomplete content; Loss: availability; lesson continuity. | Availability | Mitigate: create a default fallback page for administrative pages and fallback content for the child; send diagnositic alert to server and inform adult of error; Document/Test: test the system response with incorrectly formatted content or a mock unresponsive server; verify that fallback content replaces the returned content and that the adult user is informed | G.4, SAF-SHA-004 |

| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| H-SHA-005 | AI refinement output; feedback and guidance content | The AI refinement provider returns inaccurate or inappropriate, content which is shown to a child or adult. | The provider is prompted with guardrails that are too permissive; system error on the providers end. | A child is shown confusing or meaningless guidance or inappropriate text; Loss: well-being; trust in the application. | Psychological/well-being | Eliminate: allow users to report inappropriate output; catalogue confirmed incorrect output as part of API guardrails for output validation; never display raw provider output without filtering through validation; Document/Test: deliberately attempt to recreate inappropriate output by imitating report diagnostics; have human confirmation of reported content and adjust content validation guardrails. | G.1, G.4, SAF-SHA-005 |

| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|

## 5.5 Sensory Load, Feedback, Progress Views and Refinement (H-MOU)

Table 8 covers hazards from how Squigglr looks, sounds and speaks to the child, how its progress views describe the child to adults, and how it changes the child's creative drawings. These hazards arise even when input, saving and

Table 8: FMEA: sensory load, feedback, progress views and refinement

| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| H-MOU-001 | Child-facing screens; audio feedback | Screens or feedback present more sound, colour, movement or flashing than the child can manage. | Decorative animation, repeated or loud sounds, bright colours or many controls; children react differently to sensory input (SRC-009.05; FACT-003); no adult may be watching (CA-001). | The child becomes overwhelmed or distressed and stops practising; rapid flashing can also trigger seizures in rare cases. Loss: well-being; physical safety; the practice benefit. | Psychological/ well-being; Physical | Eliminate: no flashing on any child-facing screen, and no animation while the child draws. Mitigate: let an adult turn audio off for each child, let long sounds be paused, and keep a calm style reviewed by adults including an educator or occupational therapist. Document/Test: inspect every child-facing screen against the calm-screen checklist. | SAF-MOU-003, NFR-UX-001, NFR-UX-006, NFR-UX-007, NFR-UX-010, NFR-UX-022 |
| H-MOU-002 | Feedback wording and style | Feedback tells the child that the drawing is wrong or bad, or uses error styling, after a missed part or a non-passing attempt. | Messages written around errors; warning colours, crosses or alarm sounds; wording not yet reviewed with educators or therapists (FACT-012). | The child feels they failed, loses confidence and avoids drawing (SRC-009.02). Loss: well-being; motivation to practise. | Psychological/ well-being | Eliminate: no error styling and none of the words “wrong”, “bad”, “fail” or “mistake” in any message to the child. Mitigate: acknowledge effort before showing what to change, and review wording with an educator or occupational therapist. Document/Test: check every approved message against the word list. | NFR-UX-004, NFR-UX-007, NFR-UX-015, NFR-UX-016, FR-LRN-022 |

| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| H-MOU-003 | Feedback presentation | The child cannot tell what the feedback means or what to do next. | Feedback that needs reading, long messages, sound-only cues while audio is off, or a missed part shown only by colour to a child who does not see that colour. | The child repeats the same difficulty, gives up or needs an adult, and evaluable attempts are used up. Loss: the practice benefit; well-being. | Psychological/ well-being | Mitigate: show every message on screen and never by sound alone; keep messages short; use images instead of text; mark missed parts by more than colour. Document/Test: review all feedback with audio off and with colour hidden. | NFR-UX-002, NFR-UX-004, NFR-UX-015, NFR-UX-021 |
| H-MOU-004 | Feedback; guidance stages | The child comes to rely on Squigglr's help and does not practise drawing on their own. | Feedback completes the missed part for the child instead of pointing to it; prompts at every attempt; guidance that never fades. | The child does not move towards independent drawing, which is the aim of the practice (G.2). Loss: the practice benefit. | Psychological/ well-being | Eliminate: in guided lessons, never draw a missed part for the child. Mitigate: point to the missed part and let the child draw it; fade guidance automatically on a pass; keep the child's actual strokes. Document/Test: check that no feedback adds strokes to an attempt. | SAF-MOU-001, FR-LRN-014, FR-LRN-023, BR-004 |

| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| H-MOU-005 | Progress summaries; attempt records | A progress view makes the child look more or less independent than they were. | Assisted attempts shown as independent; refined creative drawings mixed into lesson records; fewer attempts presented as better skill (SRC-005, Moustafa, 6:03). | An adult moves the child on too early or holds them back, or overrates their progress. Loss: suitable support decisions; the child's well-being. | Data integrity; Psychological/ well-being | Eliminate: never include creative-space drawings in lesson records or summaries. Mitigate: show assisted and independent attempts separately and never present the number of attempts as a measure of skill; use plain, non-clinical wording. Document/Test: check summaries of prepared histories with known assisted and independent attempts. | SAF-MOU-002, G.6, FR-LRN-012, BR-005, SAF-RAY-004, NFR-UX-017 |
| H-MOU-006 | Creative-space refinement | Refinement changes the drawing too much: points move further than intended, or parts are added or removed. | `REFINE_TOLERANCE` too large or not enforced; the refinement service returns a different drawing; the child cannot find how to restore the original (CA-001). | The child no longer sees the drawing as their own and is upset; work changed unexpectedly is a source of frustration (SRC-009.01; SRC-009.02). Loss: well-being; trust in the creative space. | Psychological/ well-being | Mitigate: limit every change to `REFINE_TOLERANCE`; let the child or the adult restore the original or turn refinement off; let the adult turn refinement off for each child. Document/Test: run the G.4 measurement on prepared adult-made drawings. | G.4, BR-005, NFR-UX-011 |

## 5.6 Creative Drawing, Attempt Records and Summaries (H-MUH)

Table 9 covers hazards from creative drawing and its refinement, attempt records and progress summaries. Each group of rows starts with the ID and name of its component, and the Component cell repeats the name exactly. The last column cites SRS requirements by ID: creative drawing (FR-CRT), attempt record (FR-REC) and progress summary (FR-PRG) requirements. Where a row has several causes, each cause is labelled (a), (b) and so on, and the recommended action and mitigating requirements for that cause carry the same label, so that every cause traces to the requirements that mitigate it. Table 10 places the rows of all four tables under the components.

Table 9: FMEA: creative drawing, attempt records and progress summaries

| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| **C4 Creative refinement with an external model** | | | | | | | |
| H-MUH-001 | Creative refinement with an external model | A creative change cannot be reversed: undo does not step back an edit, a refinement or a restore; restoring the original is not available to the child or to the adult; or a restore does not return the original strokes exactly. | (a) The original strokes are replaced by the refined drawing, or are lost when the drawing is saved, closed and reopened. (b) A refinement or a restore is not one undoable change, or an undone change cannot be redone. (c) After several refinements, it is unclear which original a restore returns. (d) Only one of the child and the adult can restore, or the adult cannot reach the child's drawings. | The child loses their own drawing and is left with a version they did not draw, which can upset a child who wants the work to stay theirs (BR-005). Loss: L4 (data); L2 (well-being). | Data integrity; Psycho-logical/ well-being | (a) Eliminate: keep the original strokes of every refinement with the drawing, also after it is saved and reopened. (b) Mitigate: make every edit, refinement and restore one undoable change, and let an undone change be redone. (c) Mitigate: on restore, return exactly the drawing as it was before its latest refinement that has not been undone or restored. (d) Mitigate: let the child restore the original directly, and the adult from the adult's own access, after the adult check of SAF-RAY-001. Document/Test: on prepared adult-made drawings, refine, edit, close, reopen and restore, and compare the restored strokes point by point with the original. | (a) FR-CRT-017, FR-CRT-024 (b) FR-CRT-007, FR-CRT-008, FR-CRT-013 (c) FR-CRT-017 (d) FR-CRT-018, FR-CRT-019 |

*Continued on the next page*

| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| H-MUH-002 | Creative refinement with an external model | A refinement changes the drawing wrongly: a point moves further than `REFINE_TOLERANCE`; a stroke is added, removed or recoloured; only part of a result is applied; or a refinement reaches a drawing outside the supported set, an older version of the drawing, or strokes drawn in a guided lesson. | (a) The external model returns invalid, malformed, partial or excessive changes, and the team does not train it (LC-004). (b) A slow connection (CA-003) delivers a result after the child has kept drawing or has closed the drawing. (c) A stroke is refined again each time the child finishes, so the changes add up. (d) The drawing is empty or is not of a supported item (OI-03). (e) Nothing bars refinement from guided lessons. | The drawing is no longer what the child drew, and the child may no longer see it as their own; a refined guided attempt would make its record show strokes the child did not draw (BR-004). Loss: L4 (data); L2 (well-being); L5 (misleading record). | Data integrity; Psycho-logical/ well-being | (a) Mitigate: change only the paths of existing strokes and keep every refined stroke within `REFINE_TOLERANCE`; apply a result whole or not at all, and leave the drawing unchanged when refinement fails or its result breaks these rules. (b) Mitigate: discard a result computed for an earlier version of the drawing. (c) Mitigate: refine only when the child finishes a drawing, and refine a stroke again only if the child has edited it. (d) Mitigate: leave such a drawing unchanged. (e) Eliminate: refine no stroke drawn in a guided lesson. Document/Test: test with simulated invalid, excessive, partial and late results, with drawings outside the supported set, and with guided-lesson strokes. | (a) FR-CRT-011, FR-CRT-012, FR-CRT-020 (b) FR-CRT-021 (c) FR-CRT-010, FR-CRT-014 (d) FR-CRT-015 (e) FR-CRT-027 |

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| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| H-MUH-006 | Creative refinement with an external model | A refinement that meets every rule is still not accepted by the child as their own drawing: the child erases or redraws the refined parts, stops drawing, or is upset after the refinement. | (a) Refinement happens with no action from the child (BR-005), and the value children place on it has not been checked (ASM-011, FACT-013). (b) A child who cannot read may not see what changed, or that the original can come back (FACT-007). (c) Some children are agitated by unexpected changes to their work (PER-1). | The child loses the sense that the work is their own, which the creative space exists to give, and may avoid the creative space or come to see their unrefined drawings as not good enough. Loss: L2 (well-being). | Psycho-logical/ well-being | (a) Mitigate: keep every refinement small, within `REFINE_TOLERANCE`, and limited to drawings of supported items; let the child or the adult turn refinement off. Document/Test: check ASM-011 against the specialists' questionnaire answers (SRC-009.01), which do not establish children's or caregivers' preferences, and have supporting adults judge, on adult-made drawings, whether each refined drawing is still the same drawing. (b) Mitigate: show whether the open drawing holds a refinement. (c) Mitigate: let the child restore the original directly, without an adult. | (a) FR-CRT-012, FR-CRT-015, FR-CRT-030 (b) FR-CRT-016 (c) FR-CRT-018 |

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| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| **C5 Attempt storage and history** | | | | | | | |
| H-MUH-003 | Attempt storage and history | Records no longer show which work was the child's own and which was assisted: an attempt's stage is missing or wrong, a guided attempt keeps a corrected shape instead of the child's strokes, a creative drawing appears as an attempt, or a refined drawing cannot be told from the child's original. | (a) The stage is not saved with the attempt, or is taken from the lesson's stage when the save completes rather than when the child drew. (b) Strokes are smoothed or corrected before they are saved (BR-004). (c) Creative drawings and attempts are stored or counted together. (d) The originals or the refined state of a creative drawing are not saved with it. (e) A summary merges attempts made with and without a guide, or shows stages before an adult-chosen starting stage (UC-004) as failed. | A summary counts attempts made with a guide as made without one, or the reverse, so the adult moves the child on too early or holds the child back; an adult takes refined work for the child's unaided drawing. Loss: L5 (misleading information). | Data integrity | (a) Mitigate: keep the stage in every attempt record, and never change a saved record. (b) Mitigate: keep the strokes as captured, in the order drawn. (c) Eliminate: never count a creative drawing as an attempt. (d) Mitigate: save the originals of every refinement with the drawing, show whether an open drawing is refined, and restore the exact original. (e) Mitigate: keep attempts made with and without a guide apart in every summary, and mark stages not attempted. Document/Test: test summaries on prepared histories that mix both kinds of attempt, start at a later stage, and include refined creative drawings. | (a) FR-REC-001, FR-REC-009 (b) FR-REC-002 (c) FR-CRT-028 (d) FR-CRT-016, FR-CRT-017, FR-CRT-023 (e) FR-PRG-003, FR-PRG-010 |

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| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| H-MUH-004 | Attempt storage and history | An attempt record, part of one, or a creative drawing is saved to, merged with or shown under the wrong attempt, lesson record or learner profile. | (a) A save retried after a lost connection (CA-003) is stored twice or under another lesson record, or the same learner has two sessions open at once. (b) An adult switches learner or ends the session while an attempt is in progress. (c) A creative drawing is saved to the learner active when the save completes rather than when it was drawn. (d) Combined with H-RAY-006: the wrong learner is active on a shared device (CA-004); that cause and its mitigations are analysed in H-RAY-006. | One child's history holds another child's work or duplicate attempts, so counts and summaries mislead the adult; a child may see another child's drawings. Loss: L4 (data); L5 (misleading information); L3 (privacy). | Data integrity; Privacy/ security | (a) Mitigate: save each attempt record exactly once, under the lesson record of the lesson in which it was made, also after a retry and with two sessions at once; never change a saved record. (b) Mitigate: keep an attempt stopped this way as interrupted, with no outcome. (c) Mitigate: save each creative drawing to the learner who was active when it was drawn, and show only the active learner's drawings. (d) Mitigate: limit each summary to one learner, so that a mix-up is not hidden in a combined view. Document/Test: test retried saves, two sessions for one learner at once, and a learner switch during an attempt. | (a) FR-REC-008, FR-REC-009 (b) FR-REC-005 (c) FR-CRT-023, FR-CRT-026 (d) FR-PRG-001 |

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| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| H-MUH-005 | Attempt storage and history | A creative drawing, or the latest changes to it, is not saved, so it is missing, or older than the child left it, when it is reopened. | (a) The child leaves the creative space, or an adult ends the session, before the drawing is saved. (b) The connection is lost while the child draws (CA-003). (c) The page is closed while changes are still unsaved, for example while the connection is lost. | The child's work is lost; a child who wanted to finish the drawing or show it to someone is disappointed or upset. Loss: L4 (data); L2 (well-being). | Data integrity; Psycho-logical/ well-being | (a) Mitigate: save every creative drawing, with the originals of its refinements, to the active learner with no action from the child, and reopen it exactly as last saved. (b) Mitigate: save the changes made while the connection was lost once it returns. (c) Mitigate: show whether the open drawing's changes are all saved. Document/Test: test closing the page during and after a lost connection, and record which changes each reopened drawing keeps. | (a) FR-CRT-023, FR-CRT-024 (b) FR-CRT-025 (c) FR-CRT-023, FR-CRT-025, FR-CRT-029 |

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| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| H-MUH-007 | Attempt storage and history | An attempt record or lesson record does not hold what happened: a predefined difficulty is missing because the child later corrected it, the adult-set values shown are not those in use at the time, a lesson stopped early has no lesson record, or a saved record is later changed. | (a) A later correction replaces the record of the difficulty it corrected. (b) Records refer to the learner's current adult-set values, which an adult may change at any time (BR-008). (c) A lesson's progress is saved only when the lesson ends at its limit, the one case BR-002 states, and not when an adult stops it early (ASM-015). (d) Saved records are recomputed when the lesson catalogue (LC-001) or a value changes. | The adult misses a recurring difficulty, or judges a stage against the wrong pass value, and makes a poor decision about the child's support; earlier practice is lost from the history. Loss: L5 (misleading information); L4 (data). | Data integrity | (a) Mitigate: keep every predefined difficulty observed in an attempt, each marked as corrected or not. (b) Mitigate: keep with each attempt the adult-set values in use when it was made. (c) Mitigate: keep a lesson record for every lesson start, marked as unfinished when the lesson stops early. (d) Mitigate: never change a saved record except by deletion. Document/Test: compare saved records with prepared adult-made attempts that include a difficulty corrected later in the attempt (G.5). | (a) FR-REC-003 (b) FR-REC-004 (c) FR-REC-006, FR-REC-007 (d) FR-REC-009 |

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| Hazard ID | Component | Failure mode | Cause(s) | Effect and loss | Category | Recommended action | Mitigating requirement IDs |
|---|---|---|---|---|---|---|---|
| **C6 Adult access and summaries** | | | | | | | |
| H-MUH-008 | Adult access and summaries | A progress summary does not agree with the saved records behind it: a count, a stage or a recurring difficulty is wrong or missing; the latest attempts are left out without notice; interrupted attempts or stages not attempted are counted as unsuccessful; an empty history shows figures; or a figure opens attempts other than those it was computed from. | (a) The summary is computed from out-of-date or incomplete records, for example while attempts are still unsaved (CA-003). (b) Interrupted attempts and stages never attempted have no state of their own and fall into "unsuccessful", and an empty history shows figures. (c) "Recurring" has no fixed threshold. (d) A figure is not linked to the records it was computed from. | The adult misjudges the child's progress: moves the child on too early, holds the child back, or worries about failures that did not happen. Loss: L5 (misleading information). | Data integrity | (a) Mitigate: state on each summary the latest attempt it includes, and show each lesson's records in the order started, with how each lesson ended. (b) Mitigate: show interrupted attempts and stages not attempted apart from unsuccessful attempts, and state when no practice is recorded. (c) Mitigate: list a difficulty as recurring only when it is observed in at least `RECUR_MIN` attempts at one stage of one lesson. (d) Mitigate: let the adult open the attempt records behind every figure and see every value they hold. Document/Test: compare each summary with figures computed by hand for prepared histories, including empty, interrupted-only and later-starting-stage histories. | (a) FR-PRG-002, FR-PRG-011 (b) FR-PRG-008, FR-PRG-009, FR-PRG-010 (c) FR-PRG-004 (d) FR-PRG-005, FR-PRG-006 |

## 5.7 Coverage of Components and Categories

Table 10 checks the FMEA against every pair of a component and a category of harm. A hazard ID in plain type is a row of Table 5, Table 6, Table 8 or Table 9 placed under that component; an ID in parentheses is a row placed under another component whose causes lie partly in this one. The Component cell of an H-RAY, H-AHM or H-MOU row names the functions it concerns in its author's words, so this table assigns each of those rows to a component of the System Boundaries and Components section. Each ID appears in the column of its row's category; a row whose Category cell names two categories appears in both. A dash marks a pair for which no row records a failure mode.

Table 10: Coverage of the FMEA: components against categories of harm

| **Component** | **Physical** | **Psycho-logical/ well-being** | **Privacy/ security** | **Data integrity** | **Availability** |
|---|---|---|---|---|---|
| C1 Touch and stylus canvas | H-MOU-001 | H-MOU-001, H-MOU-003, (H-MOU-002), (H-MUH-001) | — | H-RAY-002, H-AHM-001, H-AHM-002, H-AHM-003, (H-MUH-001) | H-RAY-002 |
| C2 Lesson and assessment logic | — | H-RAY-004, H-RAY-005, H-MOU-002, H-MOU-004, (H-RAY-001), (H-RAY-007) | — | H-AHM-004, H-AHM-005, (H-AHM-001), (H-AHM-002), (H-AHM-003), (H-AHM-007) | — |
| C3 Guidance-suggestion logic | — | H-AHM-006 | (H-RAY-006) | — | — |
| C4 Creative refinement with an external model | — | H-MOU-006, H-MUH-001, H-MUH-002, H-MUH-006 | — | H-MUH-001, H-MUH-002, (H-MUH-003) | — |
| C5 Attempt storage and history | — | H-MUH-005, (H-RAY-003), (H-RAY-005), (H-MOU-005) | H-MUH-004, (H-RAY-006) | H-MUH-003, H-MUH-004, H-MUH-005, H-MUH-007, (H-RAY-002), (H-MOU-005) | (H-RAY-002) |
| C6 Adult access and summaries | — | H-RAY-001, H-RAY-003, H-MOU-005 | (H-MUH-004) | H-MOU-005, H-MUH-008, (H-MUH-003), (H-MUH-004) | — |
| C7 Hosting and authentication | — | — | — | — | — |
| C8 Learner profiles and adult-set values | — | H-RAY-007 | H-RAY-006, (H-MUH-004) | H-AHM-007, (H-MUH-004) | — |

Several pairs were examined and no credible failure mode was found in them:

- *Physical:* Squigglr controls no equipment, so it has two paths to bodily harm. Content that flashes on a child-facing screen can trigger a seizure; the canvas (C1) presents every child-facing screen and its feedback, so

H-MOU-001 is placed there. Practice that does not end or that restarts at once lies in the lesson and assessment logic (C2), where H-RAY-004 and H-RAY-005 end each lesson at its limit and show restarts to the adult. The guidance-suggestion logic (C3), creative refinement (C4), storage (C5), summaries (C6), hosting and authentication (C7) and learner profiles (C8) neither present child-facing content nor lengthen practice.

- *Psychological/well-being:* the child never signs in, so a failure of hosting or authentication (C7) reaches only the adult, as lost availability.
- *Privacy/security:* the canvas (C1) shows only the open attempt or drawing, and the lesson and assessment logic (C2) holds only the current attempt; showing them for the wrong learner is analysed under learner profiles (H-RAY-006) and storage (H-MUH-004).
- *Availability:* when summaries (C6) or learner profiles (C8) cannot be used, the cause is the device (H-RAY-002) or hosting and authentication (C7), so the failure is analysed there; H-RAY-002 is also listed under the canvas (C1) and storage (C5). A go-back suggestion that is not made (C3) changes nothing, and practice continues (G.3).

# 6 Safety and Security Requirements

This section lists the safety and security requirements derived from the hazard analysis (HA). A requirement is HA-derived when its ID appears in the Mitigating requirement IDs column of any of the four per-author FMEA tables (H-RAY, H-AHM, H-MOU and H-MUH) in the Failure Mode and Effect Analysis section, so that it carries out the recommended action of at least one hazard. As the teaching assistant (TA) advised (SRC-006), each requirement is written once, in the SRS, together with its rationale, fit criterion, priority, phase and sources. This section therefore adds no requirement of its own: Table 11 repeats the statement of each HA-derived requirement exactly as the SRS states it.

The table has three groups. Functional requirements come first, in the order of the SRS subsections that hold them: the Learning Activity, Creative Drawing, Attempt Record and Progress Summary requirements. The look and feel and usability requirements (NFR-UX) follow, in ID order. Safety-critical requirements come last (SAF-RAY, SAF-AHM and SAF-MOU); the SRS keeps them in its Safety-Critical Requirements subsection because they do not fit another section. For each requirement, the table gives its ID, with the short title the SRS gives it where there is one, its statement, the hazards whose FMEA rows cite it, and the SRS subsection that holds it. The subsection is named in words, because the SRS is a separate document; for the same reason, a statement that refers to an SRS table names the table by its caption. Names in monospaced capitals, such as `STAGE_PASS`, are symbolic constants that the SRS defines in its table “Central symbolic constants register”.

Some rows of the H-RAY and H-MOU tables also cite goals (G.2 to G.6), business rules (BR-004 and BR-005) or constraints (CON-001, CON-002, CON-003 and CON-007) among their mitigations. A goal states what Squigglr is for, a business rule governs who may do what with Squigglr whatever its design, and a constraint is imposed on the project from outside, so none of them is a requirement and none is repeated here. Every row that cites a goal, a business rule or a constraint also cites at least one requirement of Table 11.

Table 11: HA-derived requirements, as stated in the SRS

| ID | Requirement | Hazards | SRS section |
|---|---|---|---|
| **Functional requirements** | | | |
| FR-LRN-007 *Stylus stroke acquisition* | Squigglr shall accept stylus-drawn strokes as ordered sequences of input points for the active lesson attempt and preserve their order and stroke boundaries for subsequent classification. | H-AHM-001, H-AHM-002, H-AHM-003 | Learning Activity |
| FR-LRN-008 *Multi-stroke attempts* | Squigglr shall evaluate the collected strokes of a single attempt together rather than classify each stylus lift as a completed attempt. | H-AHM-001, H-AHM-002 | Learning Activity |
| FR-LRN-009 *Interrupted attempt handling* | Squigglr shall distinguish an interrupted or unfinished drawing attempt from a completed drawing attempt, shall not record it as passed, and shall not consume an evaluable attempt from the stage limit. | H-AHM-001, H-AHM-006 | Learning Activity |
| FR-LRN-010 *Invalid and ambiguous input* | Squigglr shall withhold passing status from an attempt whose required stylus input is missing, invalid or too ambiguous to evaluate against the stage criteria, and shall not count such input toward the stage's evaluable-attempt or unsuccessful-attempt limits. | H-AHM-001, H-AHM-003 | Learning Activity |
| FR-LRN-011 *Completeness classification* | Squigglr shall classify every evaluable drawing attempt as complete or incomplete relative to the selected lesson and guidance stage. | H-AHM-004, H-AHM-005 | Learning Activity |
| FR-LRN-012 *Guidance classification* | Squigglr shall classify each evaluated attempt as assisted when performed with outline or endpoint guidance and independent when performed without those guides. | H-MOU-005 | Learning Activity |
| FR-LRN-013 *Pass classification* | Squigglr shall report an evaluable attempt as passing its stage only when the drawing is classified as complete and its result meets the current learner's `STAGE_PASS` criterion. | H-AHM-004, H-AHM-005 | Learning Activity |

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| ID | Requirement | Hazards | SRS section |
|---|---|---|---|
| FR-LRN-014 *Predefined difficulty feedback* | Squigglr shall present only pre-approved, stage-relevant feedback observations for evaluated attempts, including applicable missed tracing portions, incomplete lines or shapes and incorrect endpoints. | H-AHM-005, H-MOU-004 | Learning Activity |
| FR-LRN-016 *Adult lesson and starting-stage selection* | Squigglr shall use the lesson and starting guidance stage selected by the supporting adult for the learner's session, including an adult-selected later starting stage. | H-AHM-007 | Learning Activity |
| FR-LRN-017 *Adult-selected practice parameters* | Squigglr shall apply the learner's adult-selected values for `STAGE_PASS`, `DIFFICULTY_STREAK`, `ATTEMPT_LIMIT`, `IMPROVE_WINDOW`, `ATTEMPT_EXTENSION` and `ATTEMPT_LIMIT_MAX`, using the documented midpoint default for any unset value and rejecting out-of-range or inconsistent selections. | H-AHM-007 | Learning Activity |
| FR-LRN-018 *Adult authorization for discretionary changes* | Squigglr shall require explicit authorization from the supporting adult before changing to another lesson, skipping a guidance stage or moving backward to an earlier stage; routine pass-based advancement to the next stage of the current lesson requires no additional adult action. | H-AHM-006 | Learning Activity |
| FR-LRN-019 *Difficulty suggestion* | Squigglr shall offer, without applying, an optional suggestion to return to an earlier drawing stage when the learner reaches `DIFFICULTY_STREAK` consecutive non-passing evaluable attempts at an eligible stage and the improvement rule does not suppress suggestions. When an adult declines or ignores the suggestion, another suggestion shall require a new full failure streak. | H-AHM-006, H-AHM-007 | Learning Activity |
| FR-LRN-020 *Improvement-based extension* | Squigglr shall suppress a difficulty suggestion while the learner qualifies as improving, and, when the unsuccessful-attempt count first reaches `ATTEMPT_LIMIT`, grant up to `ATTEMPT_EXTENSION` additional evaluable stage attempts if the average result of the most recent `IMPROVE_WINDOW` attempts exceeds the average of the preceding equally sized window by at least `IMPROVE_MARGIN`, subject to `ATTEMPT_LIMIT_MAX`. The extension shall be granted at most once per stage and expire after the allotted additional evaluable attempts, even if results continue to improve. | H-AHM-006 | Learning Activity |

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| ID | Requirement | Hazards | SRS section |
|---|---|---|---|
| FR-LRN-021 *Stage attempt limit and lesson end* | Squigglr shall end the active lesson when its unsuccessful-attempt count reaches `ATTEMPT_LIMIT` without a qualifying extension, or a granted extension is exhausted, or the allowed total-attempt count reaches `ATTEMPT_LIMIT_MAX`, and make the furthest stage reached available for the lesson record. | H-AHM-007 | Learning Activity |
| FR-LRN-022 *No learner evaluation or diagnosis* | Squigglr shall express lesson feedback and completion results as properties of the drawing attempt rather than as assessments, diagnoses or comparisons of children. | H-MOU-002 | Learning Activity |
| FR-LRN-023 *Automatic within-lesson advancement* | Squigglr shall automatically advance from full-outline tracing to endpoint-only guidance, and from endpoint-only guidance to independent drawing, upon a passing attempt at the active stage; a passing independent drawing attempt completes the current lesson without starting a different lesson. | H-AHM-004, H-MOU-004 | Learning Activity |
| FR-CRT-007 *Undo a change* | Squigglr shall let the child undo the changes to the open creative drawing one at a time, most recent first, back to the drawing as it was when the child opened it. | H-MUH-001 | Creative Drawing |
| FR-CRT-008 *Redo an undone change* | Squigglr shall let the child redo undone changes one at a time until the child makes a new change. | H-MUH-001 | Creative Drawing |
| FR-CRT-010 *Refine a finished drawing* | Squigglr shall refine the open creative drawing automatically each time the child finishes it while refinement is turned on, and at no other time. | H-MUH-002 | Creative Drawing |
| FR-CRT-011 *Change only stroke paths* | Squigglr shall change only the path of each existing stroke when it refines a creative drawing. | H-MUH-002 | Creative Drawing |
| FR-CRT-012 *Stay within the refinement tolerance* | Squigglr shall keep every refined stroke within `REFINE_TOLERANCE` of the corresponding original stroke. | H-MUH-002, H-MUH-006 | Creative Drawing |
| FR-CRT-013 *Keep the refined drawing editable* | Squigglr shall let the child edit each stroke of a refined drawing with the same actions, and with the same results, as a stroke the child drew. | H-MUH-001 | Creative Drawing |
| FR-CRT-014 *Do not refine a stroke twice* | Squigglr shall not refine again a stroke that an earlier refinement changed, unless the child has edited that stroke since. | H-MUH-002 | Creative Drawing |

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| ID | Requirement | Hazards | SRS section |
|---|---|---|---|
| FR-CRT-015 *Leave unsupported drawings unchanged* | Squigglr shall leave a finished creative drawing unchanged if it has no strokes or is not a drawing of an item in the supported set. | H-MUH-002, H-MUH-006 | Creative Drawing |
| FR-CRT-016 *Show whether a drawing is refined* | Squigglr shall show, while a creative drawing is open, whether it holds a refinement that has not been undone or restored. | H-MUH-003, H-MUH-006 | Creative Drawing |
| FR-CRT-017 *Restore the exact original* | Squigglr shall return a creative drawing, on restore, exactly to the original of its most recent refinement that has not been undone or restored. | H-MUH-001, H-MUH-003 | Creative Drawing |
| FR-CRT-018 *Child restores the original* | Squigglr shall let the child restore the original of the open creative drawing. | H-MUH-001, H-MUH-006 | Creative Drawing |
| FR-CRT-019 *Adult restores the original* | Squigglr shall let the supporting adult, after passing the adult check of SAF-RAY-001, restore the original of any creative drawing saved under a learner profile in that adult's account. | H-MUH-001 | Creative Drawing |
| FR-CRT-020 *Leave the drawing unchanged when refinement fails* | Squigglr shall make no change to a creative drawing when its refinement fails or the refinement result violates FR-CRT-011 or FR-CRT-012. | H-MUH-002 | Creative Drawing |
| FR-CRT-021 *Discard an out-of-date result* | Squigglr shall discard a refinement result if the drawing has been changed or closed since that refinement started. | H-MUH-002 | Creative Drawing |
| FR-CRT-023 *Save to the active learner* | Squigglr shall save each creative drawing, with the originals of its refinements, to the learner profile that was active when the child drew it, without any action from the child. | H-MUH-003, H-MUH-004, H-MUH-005 | Creative Drawing |
| FR-CRT-024 *Reopen a drawing as it was left* | Squigglr shall reopen a saved creative drawing exactly as it was last saved, with every stroke editable and every refinement that has not been undone or restored still restorable. | H-MUH-001, H-MUH-005 | Creative Drawing |
| FR-CRT-025 *Keep changes through a lost connection* | Squigglr shall save, once the connection returns, every change the child made to an open creative drawing while the connection was lost. | H-MUH-005 | Creative Drawing |
| FR-CRT-026 *Show only the active learner's drawings* | Squigglr shall show in the creative space only the creative drawings saved under the active learner profile. | H-MUH-004 | Creative Drawing |

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| ID | Requirement | Hazards | SRS section |
|---|---|---|---|
| FR-CRT-027 *No refinement in guided lessons* | Squigglr shall not refine any stroke drawn in a guided lesson. | H-MUH-002 | Creative Drawing |
| FR-CRT-028 *Creative drawings are not attempts* | Squigglr shall not count a creative drawing as an attempt. | H-MUH-003 | Creative Drawing |
| FR-CRT-029 *Show unsaved changes* | Squigglr shall show, while a creative drawing is open, whether every change to it has been saved. | H-MUH-005 | Creative Drawing |
| FR-CRT-030 *Turn refinement off* | Squigglr shall let the child, and the supporting adult, turn automatic refinement off and on for a learner profile. | H-MUH-006 | Creative Drawing |
| FR-REC-001 *Attempt record* | Squigglr shall keep, for each attempt that ends with an outcome, an attempt record holding the lesson, the stage, the attempt's result, the outcome (whether the attempt passed its stage, FR-LRN-013) and the date and time the attempt ended. | H-MUH-003 | Attempt Record |
| FR-REC-002 *Actual strokes* | Squigglr shall keep in each attempt record the strokes the child drew in the attempt, in the order drawn, as they were captured. | H-MUH-003 | Attempt Record |
| FR-REC-003 *Difficulties and corrections* | Squigglr shall keep in each attempt record every predefined difficulty observed during the attempt, each marked as corrected or not corrected within that attempt. | H-MUH-007 | Attempt Record |
| FR-REC-004 *Adult-set values in use* | Squigglr shall keep in each attempt record the value of each adult-set value that was in use for the learner when the attempt was made. | H-MUH-007 | Attempt Record |
| FR-REC-005 *Interrupted attempt* | Squigglr shall keep an attempt that an adult stops in Squigglr, by ending the session or switching learner, after at least one stroke but before its outcome is decided, as an attempt record marked as interrupted, holding the strokes and predefined difficulties up to that point and no outcome. | H-MUH-004 | Attempt Record |
| FR-REC-006 *Lesson record* | Squigglr shall keep, each time a learner starts a lesson, a lesson record holding the lesson, the starting stage, the furthest stage reached, the number of successful attempts at each stage and how the lesson ended. | H-MUH-007 | Attempt Record |

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| ID | Requirement | Hazards | SRS section |
|---|---|---|---|
| FR-REC-007 *Unfinished lesson* | Squigglr shall keep the lesson record of a lesson that stops before its last stage is passed and before it reaches an attempt limit, marked as unfinished, with the furthest stage reached and the successful attempts at each stage up to that point. | H-MUH-007 | Attempt Record |
| FR-REC-008 *Each attempt saved once* | Squigglr shall save each attempt record exactly once, under the lesson record of the lesson in which the attempt was made. | H-MUH-004 | Attempt Record |
| FR-REC-009 *Saved records unchanged* | Squigglr shall keep the contents of every saved attempt record, and of every lesson record whose lesson has ended, unchanged except by deletion. | H-MUH-003, H-MUH-004, H-MUH-007 | Attempt Record |
| FR-PRG-001 *One learner per summary* | Squigglr shall show in each progress summary only the records of the one learner profile that the summary names. | H-MUH-004 | Progress Summary |
| FR-PRG-002 *Progress over time by lesson* | Squigglr shall show, for each lesson in a progress summary, the learner's lesson records in the order they were started, each with its starting stage, furthest stage reached, successful attempts at each stage and how the lesson ended. | H-MUH-008 | Progress Summary |
| FR-PRG-003 *With and without a guide kept apart* | Squigglr shall show attempts made without a guide separately from attempts made with a guide in every part of a progress summary that counts or lists attempts. | H-MUH-003 | Progress Summary |
| FR-PRG-004 *Recurring difficulties* | Squigglr shall list in each progress summary every predefined difficulty observed in at least `RECUR_MIN` of the learner's attempts at the same stage of the same lesson, with the number of those attempts and the number in which the child corrected it. | H-MUH-008 | Progress Summary |
| FR-PRG-005 *Open the attempts behind the summary* | Squigglr shall let the adult open, from each figure or item in a progress summary, the attempt records from which it was computed. | H-MUH-008 | Progress Summary |
| FR-PRG-006 *Attempt record view* | Squigglr shall show, for each attempt record the adult opens, every value kept in it under FR-REC-001 to FR-REC-005, with the strokes drawn as recorded. | H-MUH-008 | Progress Summary |
| FR-PRG-008 *Empty history* | Squigglr shall show, when the learner has no saved attempt record in the chosen lessons and dates, a progress summary that states that no practice is recorded and shows no stage, count or difficulty. | H-MUH-008 | Progress Summary |

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| ID | Requirement | Hazards | SRS section |
|---|---|---|---|
| FR-PRG-009 *Interrupted attempts shown apart* | Squigglr shall show interrupted attempts in a progress summary separately from successful and unsuccessful attempts. | H-MUH-008 | Progress Summary |
| FR-PRG-010 *Stages not attempted* | Squigglr shall mark each stage of a lesson that the learner has not attempted, including a stage before an adult-chosen starting stage, as not attempted, distinct from a stage attempted without success. | H-MUH-003, H-MUH-008 | Progress Summary |
| FR-PRG-011 *Latest attempt included* | Squigglr shall state on each progress summary the date and time of the most recent attempt it includes. | H-MUH-008 | Progress Summary |
| **Look and feel and usability requirements** | | | |
| NFR-UX-001 *Calm, uncluttered child screens* | Each child-facing screen shall offer the child no more than `CHILD_CONTROLS_MAX` controls and shall show no decorative animation or flashing while the child is drawing. | H-MOU-001 | Appearance |
| NFR-UX-002 *Images and icons instead of text* | Every control on a child-facing screen shall be identified by an image or icon, and no task on a child-facing screen shall require the child to read. | H-MOU-003 | Appearance |
| NFR-UX-004 *Feedback that never marks the drawing as wrong* | Feedback on an attempt, including a missed part or a non-passing attempt, shall use encouraging styling, shall not use error styling such as warning colours, crosses, alarm sounds or "wrong" symbols, and shall not show a missed part by colour alone. | H-MOU-002, H-MOU-003 | Style |
| NFR-UX-006 *Appealing but calm style* | Child-facing screens shall have a style that adult reviewers judge both appealing to children and calm. | H-MOU-001 | Style |
| NFR-UX-007 *Style reviewed with educators or therapists* | The colours, images and feedback style of child-facing screens shall be reviewed by at least `UX_REVIEWERS` adults, including at least one educator or occupational therapist, and each change they request shall be made or recorded with a reason. | H-MOU-001, H-MOU-002 | Style |
| NFR-UX-010 *Audio setting for each child* | The supporting adult shall be able to turn audio feedback on or off for each child, and the choice shall apply to every child-facing screen for that child. | H-MOU-001 | Personalization and Internationalization |
| NFR-UX-011 *Refinement setting for each child* | The supporting adult shall be able to turn creative-space refinement on or off for each child; while it is off, the child's strokes shall be kept exactly as drawn. | H-MOU-006 | Personalization and Internationalization |

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| ID | Requirement | Hazards | SRS section |
|---|---|---|---|
| NFR-UX-015 *Short, encouraging messages to the child* | Every message to the child, including feedback on an attempt and the end of a lesson, shall be encouraging, shall describe the drawing rather than the child, and shall be at most `FEEDBACK_WORDS_MAX` words when spoken. | H-MOU-002, H-MOU-003 | Understandability and Politeness |
| NFR-UX-016 *Effort acknowledged* | The feedback on every evaluable attempt, including one that does not pass, shall acknowledge the child's effort before it shows what to change. | H-MOU-002 | Understandability and Politeness |
| NFR-UX-017 *Plain wording in adult views* | Adult views shall use plain language, shall explain each glossary term where it first appears in a view, and shall not use diagnostic, clinical or ranking words. | H-MOU-005 | Understandability and Politeness |
| NFR-UX-021 *Feedback not by sound alone* | Every message to the child shall be shown on screen and, unless audio is turned off, also played as sound; no message shall rely on sound alone. | H-MOU-003 | Accessibility |
| NFR-UX-022 *Control of audio* | Any sound that plays automatically for more than 3 seconds shall have a control to pause or stop it. | H-MOU-001 | Accessibility |
| **Safety-critical requirements** | | | |
| SAF-RAY-001 | Squigglr shall allow lesson choice, starting-stage choice, learner switching and acceptance of a go-back suggestion only after an adult check that a child cannot complete by tapping icons, and shall keep a pending suggestion, without blocking practice, until an adult accepts or declines it. | H-RAY-001, H-RAY-006 | Safety-Critical |
| SAF-RAY-002 | Before a lesson starts, Squigglr shall check the device and browser against `SUPPORTED_DEVICES` (TBD) and, if they do not match, show an adult-readable notice that results may be unreliable. | H-RAY-002 | Safety-Critical |
| SAF-RAY-003 | Squigglr shall keep each completed attempt on the device until the server confirms it is saved, and shall tell the adult which attempts could not be saved. | H-RAY-002 | Safety-Critical |
| SAF-RAY-004 | No record, summary or export shall show a diagnostic label, a judgement of the child's abilities or a comparison with other children, and every summary and export shall state that it describes practice only and is not an assessment or diagnosis. | H-RAY-003, H-MOU-005 | Safety-Critical |

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| ID | Requirement | Hazards | SRS section |
|---|---|---|---|
| SAF-RAY-005 | Squigglr shall end a lesson at `ATTEMPT_LIMIT`, or at `ATTEMPT_LIMIT_MAX` when extended, without any action from the child, and shall keep the progress saved up to that point. | H-RAY-004 | Safety-Critical |
| SAF-RAY-006 | Squigglr shall record each lesson that ends at `ATTEMPT_LIMIT` or `ATTEMPT_LIMIT_MAX`, and each improvement extension, show the adult how many times these happened for each lesson, and keep earlier attempts when a lesson is restarted. | H-RAY-005 | Safety-Critical |
| SAF-RAY-007 | Squigglr shall identify the active learner on every practice screen in a form a child who cannot read can recognize, and shall save every attempt and suggestion to that learner only. | H-RAY-006 | Safety-Critical |
| SAF-RAY-008 | Squigglr shall save a child's adult-set values (the SRS table "Central symbolic constants register") only within their team-set ranges, with `ATTEMPT_LIMIT` larger than `DIFFICULTY_STREAK` and `ATTEMPT_LIMIT_MAX` at least `ATTEMPT_LIMIT`; shall use the middle of each range when the adult does not choose; shall allow changes only after the adult check; and shall show the values in use alongside the child's records. | H-RAY-007 | Safety-Critical |
| SAF-AHM-001 | Squigglr shall prevent duplicate reports of the same stylus input event from being counted more than once during evaluation or recording of an attempt, while preserving legitimate separately drawn strokes, including overlapping strokes. | H-AHM-002 | Safety-Critical |
| SAF-AHM-002 | Squigglr shall preserve the relative positions and order of stylus points within the active drawing surface when the canvas size or supported device orientation changes, and shall reject input whose coordinates cannot be mapped reliably to that surface. | H-AHM-003 | Safety-Critical |
| SAF-MOU-001 | In guided lessons, feedback on a missed part shall show the child where the part is missing without drawing it for them. | H-MOU-004 | Safety-Critical |
| SAF-MOU-002 | Records and progress summaries shall keep attempts made with a guide separate from attempts made without one, shall not include creative-space drawings, and shall not present the number of attempts as a measure of the child's skill. | H-MOU-005 | Safety-Critical |

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| ID | Requirement | Hazards | SRS section |
|---|---|---|---|
| SAF-MOU-003 | No content on any child-facing screen shall flash more than three times in any one-second period. | H-MOU-001 | Safety-Critical |

In total, 82 requirements are HA-derived: 16 Learning Activity, 22 Creative Drawing, nine Attempt Record and ten Progress Summary requirements, five Look and Feel and seven Usability and Humanity requirements, and 13 safety-critical requirements. Every row of the four FMEA tables cites at least one of them. The Roadmap section that follows states the phase in which each is planned.

# 7 Roadmap

This section states when each HA-derived requirement of Table 11 is planned to be met. The capstone timeline has three phases, each named after the course deliverable that ends it: PH-1, the proof-of-concept (PoC) demonstration, from Nov. 23, 2026; PH-2, Revision 0 (Rev 0), from Feb. 1, 2027; and PH-3, the final demonstration, from Mar. 15, 2027. These identifiers are those of the SRS Introduction and are distinct from the interview participant codes. Each date is the start of the course window for that deliverable in the Development Plan (SRC-002). Work planned after the capstone has the phase Future. A phase states when a requirement is planned to be met; no requirement is claimed here to be implemented. The PoC experiments use team-created drawings and prepared attempts (SRC-002), and the PoC does not need a complete application, so the requirements planned for PH-2 protect the people who use the product as a whole.

The phase of a functional, look and feel or usability requirement is the one in its Phase field in the SRS. The safety-critical requirements have no Phase field, so for each of them Table 12 gives the phase of the functions it protects and the reason. The table gives one reason for each group of requirements that share a purpose, and names the hazards each group mitigates.

Table 12: Phase of each HA-derived requirement

| **Phase (date)** | **Requirement IDs** | **Rationale** |
|---|---|---|
| PH-1 (Nov. 23, 2026) | FR-CRT-007, FR-CRT-010, FR-CRT-011, FR-CRT-012, FR-CRT-013, FR-CRT-017, FR-CRT-018 | Editable refinement that can be undone is the main technical uncertainty of the PoC. Its second experiment refines drawings of the agreed supported set into editable paths, modifies the result, undoes the refinement and compares the restored drawing with the saved original (SRC-002). These requirements state what that experiment must show, so the controls on a creative change that cannot be undone or is applied wrongly are planned from the first phase (H-MUH-001, H-MUH-002, H-MUH-003, H-MUH-006). |
| PH-1 (Nov. 23, 2026) | FR-CRT-020 | The third PoC experiment must show that an invalid instruction from a candidate model is rejected and leaves the drawing unchanged (SRC-002). This requirement states that behaviour. If the model investigation is deferred, as the Development Plan allows, the requirement still applies to a failed geometric refinement (H-MUH-002). |
| PH-1 (Nov. 23, 2026) | FR-REC-001, FR-REC-003 | The fourth PoC experiment inspects the attempt records to confirm that the observations and the assistance used remain available (SRC-002). These two requirements define that record: each attempt's lesson, stage and outcome, and every predefined difficulty observed, marked as corrected or not (H-MUH-003, H-MUH-007). |
| PH-1 (Nov. 23, 2026) | FR-LRN-007, FR-LRN-010, SAF-AHM-001, SAF-AHM-002 | The first PoC experiment draws with touch and a compatible stylus on the target tablet, repeats the interactions on an additional device and browser, and checks that strokes follow the input position and remain continuous (SRC-002). These requirements state what the capture must keep: the order and boundaries of strokes (FR-LRN-007), no passing status for missing or ambiguous input (FR-LRN-010), no repeated count of one input event (SAF-AHM-001) and no distortion of stroke positions when the canvas size or device orientation changes (SAF-AHM-002) (H-AHM-001, H-AHM-002, H-AHM-003). |
| PH-1 (Nov. 23, 2026) | FR-LRN-011, FR-LRN-013, FR-LRN-016, FR-LRN-023 | The fourth PoC experiment uses prepared sequences of successful and unsuccessful attempts and demonstrates reduced assistance following success (SRC-002). These requirements state when an attempt is complete and passes (FR-LRN-011, FR-LRN-013), that the lesson and starting stage the adult chose are used (FR-LRN-016) and that a passing attempt advances within the lesson without a further adult action (FR-LRN-023) (H-AHM-004, H-AHM-005, H-AHM-007, H-MOU-004). |
| PH-1 (Nov. 23, 2026) | FR-LRN-022, NFR-UX-004, NFR-UX-015 | These requirements set the wording and styling of every message to the child: it describes the drawing rather than the child, never marks the drawing as wrong and stays short. Their phase is the one in their Phase field in the SRS (H-MOU-002, H-MOU-003). |

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| Phase (date) | Requirement IDs | Rationale |
|---|---|---|
| PH-2 (Feb. 1, 2027) | FR-CRT-014, FR-CRT-015, FR-CRT-016, FR-CRT-019, FR-CRT-021 | Rev 0 demonstrates the product as a whole, after the focused PoC experiments. These requirements bound refinement in the cases the PoC experiments do not cover: a stroke that would be refined twice, a drawing outside the supported set, a result that arrives after the drawing has changed, a refinement that cannot be seen, and a restore by the adult after the adult check of SAF-RAY-001 (H-MUH-001, H-MUH-002, H-MUH-003, H-MUH-006). |
| PH-2 (Feb. 1, 2027) | FR-CRT-023, FR-CRT-024, FR-CRT-026, FR-CRT-027, FR-CRT-028 | Saving creative drawings to learner profiles, and keeping them apart from guided lessons and attempts, needs adult accounts, learner profiles and stored records. The PoC does not need a complete application (SRC-002), so these requirements are planned for Rev 0 (H-MUH-001, H-MUH-002, H-MUH-003, H-MUH-004, H-MUH-005). |
| PH-2 (Feb. 1, 2027) | FR-CRT-030, NFR-UX-011 | Refinement is turned off for a learner profile, so these requirements need the learner profiles and settings planned for Rev 0, like the requirements that keep creative drawings to one learner. Until then, every refinement stays within `REFINE_TOLERANCE` and can be undone or restored (FR-CRT-007, FR-CRT-012, FR-CRT-017, FR-CRT-018, PH-1) (H-MUH-006, H-MOU-006). |
| PH-2 (Feb. 1, 2027) | FR-REC-002, FR-REC-004 to FR-REC-009 | Complete attempt and lesson records, each saved once and unchanged after saving, are needed as soon as adults rely on the records (G.5). The values kept by FR-REC-004 have ranges that the SRS table "Central symbolic constants register" defines; their verification is due Nov. 2, 2026 (OI-05) (H-MUH-003, H-MUH-004, H-MUH-007). |
| PH-2 (Feb. 1, 2027) | FR-PRG-001 to FR-PRG-006, FR-PRG-008 to FR-PRG-011 | Progress summaries are built from the complete records, so they are planned with them (G.6). These requirements keep each summary to one learner and consistent with the records behind it (H-MUH-003, H-MUH-004, H-MUH-008). |
| PH-2 (Feb. 1, 2027) | SAF-RAY-001, SAF-RAY-007 | Adult-only actions, learner switching and an active learner exist only once the product has adult accounts and several learner profiles, which the PoC does not need. Both are planned with the requirements that keep drawings and summaries to one learner (FR-CRT-023, FR-CRT-026, FR-PRG-001) (H-RAY-001, H-RAY-006). |
| PH-2 (Feb. 1, 2027) | SAF-RAY-002 | The device check needs the list of supported devices and browsers (`SUPPORTED_DEVICES`), which the team sets by Nov. 2, 2026 (OI-09). The first PoC experiment records the hardware and browser versions tested (SRC-002), which informs that list (H-RAY-002). |
| PH-2 (Feb. 1, 2027) | SAF-RAY-003 | Keeping each completed attempt on the device until the server confirms the save protects the stored records, which are planned for Rev 0. The requirement is planned with FR-REC-008 and FR-PRG-011 (H-RAY-002). |

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| Phase (date) | Requirement IDs | Rationale |
|---|---|---|
| PH-2 (Feb. 1, 2027) | SAF-RAY-004, SAF-MOU-002, FR-LRN-012, NFR-UX-017 | These requirements apply to every record view and progress summary, which are planned for Rev 0 with the Progress Summary requirements. SAF-MOU-002 keeps apart the assisted and independent attempts that FR-LRN-012 classifies; FR-LRN-012 and NFR-UX-017 have this phase in their Phase field in the SRS (H-RAY-003, H-MOU-005). |
| PH-2 (Feb. 1, 2027) | SAF-RAY-005, SAF-RAY-006 | Ending a lesson at its limit, and recording lessons that reach the limit and are restarted, belong to the complete lesson flow and the lesson record. The fourth PoC experiment, as the Development Plan describes it, demonstrates changes in assistance rather than lesson ends (SRC-002). Both are planned with FR-REC-006 and FR-REC-007, and SAF-RAY-005 with FR-LRN-021, which ends the lesson at its attempt limit (H-RAY-004, H-RAY-005). |
| PH-2 (Feb. 1, 2027) | SAF-RAY-008 | Checking adult-set values uses their team-set ranges, which the SRS table "Central symbolic constants register" defines and whose verification is due Nov. 2, 2026 (OI-05). Applying the adult's values is planned with FR-LRN-017, and showing the values in use with the records with FR-REC-004 (H-RAY-007). |
| PH-2 (Feb. 1, 2027) | FR-LRN-008, FR-LRN-009 | Attempts of several strokes, and attempts that are interrupted or unfinished, extend the stroke capture planned for PH-1 (FR-LRN-007). Their phase is the one in their Phase field in the SRS (H-AHM-001, H-AHM-002, H-AHM-006). |
| PH-2 (Feb. 1, 2027) | FR-LRN-014, FR-LRN-017 to FR-LRN-021 | Feedback observations, the adult's chosen values, adult authorization of discretionary changes, the difficulty suggestion, the improvement extension and the end of a lesson at its attempt limit belong to the complete lesson flow. Their phase is the one in their Phase field in the SRS, and it matches that of SAF-RAY-005, SAF-RAY-006 and SAF-RAY-008, which protect the same functions (H-AHM-005, H-AHM-006, H-AHM-007, H-MOU-004). |
| PH-2 (Feb. 1, 2027) | SAF-MOU-001, NFR-UX-016 | Feedback on a missed part, and feedback that acknowledges the child's effort, are part of the feedback observations of FR-LRN-014. SAF-MOU-001 is planned with FR-LRN-014, and NFR-UX-016 has this phase in its Phase field in the SRS (H-MOU-002, H-MOU-004). |
| PH-2 (Feb. 1, 2027) | NFR-UX-001, NFR-UX-002, NFR-UX-010, NFR-UX-021, NFR-UX-022, SAF-MOU-003 | Child-facing screens and audio feedback are complete only in the product as a whole, which Rev 0 demonstrates; the PoC experiments use team-created drawings and prepared attempts (SRC-002). The phase of each NFR-UX requirement is the one in its Phase field in the SRS. SAF-MOU-003 is planned with NFR-UX-001, which bans flashing while the child draws (H-MOU-001, H-MOU-003). |

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| Phase (date) | Requirement IDs | Rationale |
|---|---|---|
| PH-3 (Mar. 15, 2027) | FR-CRT-008 | Redo (priority Could) only recovers a change undone by mistake. Until PH-3, every change can still be undone (FR-CRT-007, PH-1), and the original of a refinement can be restored (FR-CRT-017, FR-CRT-018, PH-1) (H-MUH-001). |
| PH-3 (Mar. 15, 2027) | FR-CRT-025 | Keeping the changes made while the connection is lost (priority Should) is the only requirement that saves them (H-MUH-005). Until PH-3, every creative drawing is saved without any action from the child and reopens as it was left (FR-CRT-023, FR-CRT-024, PH-2), so only the changes made while the connection is lost are at risk. Offline use is out of scope (ASM-005), and the FMEA's test of closing the page during a lost connection records which changes are kept (H-MUH-005). |
| PH-3 (Mar. 15, 2027) | FR-CRT-029 | Showing whether every change has been saved (priority Should) lets an adult keep the page open until a change made while the connection was lost has been saved, because offline use is out of scope (ASM-005) and such changes are kept only while the page stays open (FR-CRT-025) (H-MUH-005). |
| PH-3 (Mar. 15, 2027) | NFR-UX-006, NFR-UX-007 | Review of the style by adults, including an educator or occupational therapist (priority Should), needs the finished child-facing screens. Their phase is the one in their Phase field in the SRS. Until PH-3, child-facing screens offer few controls and show no flashing or animation while the child draws (NFR-UX-001, SAF-MOU-003, PH-2), and feedback never marks a drawing as wrong (NFR-UX-004, PH-1) (H-MOU-001, H-MOU-002). |

No HA-derived requirement is planned for Future: every row of the four FMEA tables has its mitigating requirements planned within the capstone. Four of the items left outside the scope of this analysis, as the Scope and Purpose of Hazard Analysis section explains, are features that may be added later. Each would need hazards and safety requirements of its own before it is built, so those safety requirements are Future work:

- *Camera-based finger tracing*, a stretch goal (SRC-001): a camera would capture images of the child, while a learner profile holds only a first name and a non-identifying username (BR-001), so its privacy hazards must be analysed first.
- *Letter tracing and drawing*, a stretch goal (SRC-001): the existing requirements apply to letter lessons, and letter-specific hazards, such as an adult reading letter attempts as an assessment (BR-006), need requirements when letters are specified.
- *Finger input as a supported input* (CON-003, LC-006): stage checks and value ranges are set for stylus input, so the canvas and lesson hazards must be reviewed before finger input is supported.

- *An offline mode* (ASM-005): changes and attempts kept without a connection would extend FR-CRT-025 and SAF-RAY-003, whose hazards are H-MUH-005 and H-RAY-002.

# References


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# 8 Appendix — Generative AI Use Disclosure

[ChatGPT (version 5) was used to brainstorm ideas for the template for this disclose section. —SS]

## 8.1 AI Tools Used

[Give the name of the AI tool(s) used and the version. —SS]

## 8.2 How and Where Used

[Explain what the tool(s) were used for. Examples include: brainstorming, explaining a technical concept, reviewing the document, generating or modifying text, generating or modifying diagrams, identifying errors or omissions or inconsistencies. —SS]

[For each of the uses, identify which parts of the document were affected. —SS]

[You do not need to report every interaction, but you should disclose the uses that materially contributed to your work —SS]

## 8.3 Verification of AI-Generated Content

[

- Describe how the team checked AI-generated or AI-assisted material before including it in the document.
- In particular, verify factual claims, technical assertions, requirements, calculations, references, and other information for which an AI system may produce incorrect or fabricated results.
- **The team is responsible for the accuracy, completeness, consistency, and appropriateness of the submitted document, regardless of whether material was generated by a human or an AI system.**

—SS]

### Individual Disclosures

**Rayan:** I used Anthropic's Claude Opus 5.5 at medium effort, accessed through Claude Code, while preparing my sections of this document. I developed the ideas and made the project decisions in my sections myself, and Moustafa and I conducted the specialist interview and follow-up questionnaire. A major use of AI was organizing our sources: I used the model to sort through the interview transcript, questionnaire answers, and meeting notes, and to extract the specific statements, with their timestamps, that supported the ideas I wanted to include. This would have been very difficult to do by hand given the length of

the material. I also used it to turn my notes and decisions into draft wording, to improve sentence structure, clarity, and flow, and to help with LaTeX formatting. Through Claude Code, it also helped manage the repository (branches, commits, and merges) and helped me review my teammates' pull requests by checking their changes against our decisions and the rest of the document; I decided which review comments to post.

All ideas and decisions in my sections are my own. I reviewed every AI-assisted passage against my intended meaning and checked each cited statement against the original transcript, questionnaire, or meeting notes to confirm that it says what we claim it says.

**Shamil:** For the FMEA table I used the Kiro IDE which uses automatically selects an optimal LLM for consistency among multiple versions of Claude and others. I used the chatbot to generate a skeleton of table section with starting text for idea generation and brainstorming and kept short column text where it was only simple classifiers. I mostly used my own words for the written content, but occasionally repurposed generated text with rewording, paraphrasing, or by completely rewriting. When rewriting, I removed sections based on false assumptions and made sure it was consistent with the rest of the document. None of the text for the reflection questions is AI-generated. All decisions in my sections are my own and I reviewed all AI-assisted passages thoroughly.

**Ahmed:** I used OpenAI's ChatGPT (GPT-6) while writing my parts of the Hazard Analysis, particularly the H-AHM failure modes, their effects and mitigations, and the links to the SRS. All of the ideas behind my contributions are my own, and the project decisions were made through discussion with my teammates. I used AI to help put those ideas into words, improve grammar and sentence structure, and keep the technical descriptions clear and consistent. It also helped me express the analysis in the LaTeX table format and refine wording where my hazards overlapped with a teammate's work. Our team also used AI to help divide the workload fairly and evenly among members, taking into account the scope and complexity of each task. I personally used it to help write clear and descriptive pull request descriptions that accurately summarized the changes I made, making them easier for my teammates to review.

I reviewed the AI-assisted wording before including it, checked that the failure modes and mitigations reflected what I intended, and compared the requirement references with the SRS. The final choices about what to include and how to address each risk were mine and the team's, not the model's.

**Moustafa:** I used Anthropic's Claude Opus 5.5, accessed through Claude Code, alternating between medium and high effort depending on the task. Because the document was so large, I relied on it heavily to save time on project management. Connected to the GitHub CLI, it split my assignment into ten sub-issues so I could track my work in smaller pieces, checked which tasks I still had left, and made my commits and pull requests. For writing, I decided what each part should say and gave it my ideas, sometimes after brainstorming with it. It then wrote a first draft and the LaTeX for it, which I refined until it said exactly what I intended. In this document, that covered the H-MOU rows of the FMEA and their links to the SRS. I also used it for review: as team-

mates merged their sections, I had it read the document again to check that my additions stayed consistent with the rest and contradicted nothing, and it helped me review my teammates' pull requests. GitHub Copilot also gave a light automated review on a few of my pull requests.

All ideas in my sections are my own. I reviewed every AI-drafted passage against my intended meaning before including it, and I decided which flagged inconsistencies and review comments were valid and how to resolve them.

# 9 Appendix — Reflection

[Not required for CAS 741 —SS]

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?

   **Rayan:** Identifying the hazards was easier than expected because we already had good sources to work from. The transcript of our interview with three domain specialists on October 1, 2026 gave us concrete statements, with timestamps, about how the intended children use apps and what could go wrong for them. For example, the specialists said adults should decide progression rather than the app, that audio feedback should be possible to mute, and that Squigglr may be used at home, at school or in other settings. Each of these pointed directly to a failure mode we could analyse, so we were not guessing at hazards from scratch. Our meeting with our supervisor, Dr. Anand, on October 14, 2026 helped in the same way: once Dr. Anand explained that thresholds should be set by the adult for each child within a range, the hazards around wrong or extreme threshold values became clear. Because the SRS already recorded these sources, business rules and assumptions, we could trace each hazard and critical assumption back to a specific statement, which also made the mitigations easier to justify.

   **Shamil:** Identifying mitigationa and elimination strategies for the hazards went well for me because I have previously taken classes on requirements and testing. I was able to select proper testing methods like integration testing or identify which of my original testing ideas were not fit for the project. For instances where the AI refinement provider returned inaccurate or inappropriate content (H-SHA-005), I considered fuzz testing to assess the durability and coverage of the API guardrails. After realizing that this would require human confirmation regardless, I decided to just have a human tester recreate the problematic outputs after modifying the guardrails. Similarly, I recognized that grey-box testing was suited for the case in which confidential user data is mistakenly shared with a non-essential third-party service (H-SHA-002). Testers can run different scenarios and observe the public-facing API to see if the target content is

being shared or not. The requirements course also went over testing in relations to requirements, so I was able to connect the safety requirements to specific hazards without issue.

**Ahmed:** What went well for me was having the learning activity requirements written in enough detail that I could actually think through how they might fail. I had been working on the guided-lesson flow in the SRS, so I already knew the steps from drawing a stroke to classifying an attempt, showing feedback and deciding whether the learner advances. Looking at each of those steps made it much easier to come up with specific hazards instead of just listing general problems that any website could have.

It also helped that we were working on the SRS and Hazard Analysis at the same time. When I identified a failure mode, I could check whether an existing requirement already addressed it and point to that requirement directly. For duplicate stylus events and incorrect coordinate mapping, I could see that we needed two more explicit safeguards in the SRS. I think that connection between the two documents was the strongest part of my work, because the hazards led to concrete requirements that we can eventually test.

**Moustafa:** What went well was how much this deliverable changed the way I think about what we're building. When I think of developing a drawing app for kids, the last thing that comes to mind is hazards and safety. But going through my rows made me realize how many gaps we could have had if we hadn't thought about this. For example, I never would have guessed a drawing app could cause physical harm, but rapid flashing can trigger seizures in people with photosensitive epilepsy, which is why WCAG limits flashing to three times a second. That became H-MOU-001 and SAF-MOU-003. I think thinking in terms of hazards early on is going to be really useful for my career in general, not just for this project.

2. What pain points did you experience during this deliverable, and how did you resolve them?

   **Rayan:** The main pain point was that my hazards kept changing as other parts of the documents changed. I wrote my first set of hazards and critical assumptions based on the SRS as it stood at the time, where the team would set fixed values for thresholds such as `DIFFICULTY_STREAK` and `ATTEMPT_LIMIT`. In our meeting on October 14, 2026, Dr. Anand advised that suitable values differ from child to child, so I changed those thresholds to be set by the supporting adult within a range the team defines. When I reviewed the hazard analysis again, I found that this changed the hazards: the risk was no longer only a wrong team-chosen value, but also an adult choosing a value at the edge of the range, so I added a new hazard and safety requirement for it. The same thing happened when I added the improvement rule, which can extend a lesson while a child is improving, and when we confirmed that the child does

not navigate inside a lesson. Each time, I had to go back and update the affected hazards, critical assumptions and mitigations so they matched the SRS. It was time-consuming, but it showed me how closely the hazard analysis depends on the requirements, and that it has to be reviewed again whenever the SRS changes.

**Shamil:** My main pain point was in producing FMEA items that would translate to non-overlapping requirements. I accomplished this well by making unauthorized data access (H-SHA-001) and data leakage to third parties (H-SHA-002) distinct as well as provider outage (H-SHA-004) and inappropriate provider response (H-SHA-005). The former two could have been unauthorized data access in general, however this would not cover cases in which our own service leaks data unknowingly. Splitting these cases into two rows also gave room to discuss account access in general, which would not apply to third-party data leakage. For the latter two, I decided that cases in which a provider outage sends a default or fallback data response which is erroneously used in the application does not necessarily overlap with inappropriate AI output because the first case can be fixed simply with a single error catch whereas the second case must be handled individually and updated with new incorrect outputs.

**Ahmed:** The hardest part was deciding where one hazard ended and another began. A mistake in the drawing input can lead to a wrong classification, which can then lead to incorrect feedback or an inappropriate stage change. At first it was tempting to describe all of those consequences in every row. I had to keep going back and asking what the actual failure mode was, what caused it, and which requirement would prevent or reduce it.

There was also some overlap with the hazards being written by other team members. In particular, my initial feedback hazard was too close to the work on confusing or discouraging feedback. After reviewing that feedback, I narrowed my row to a technical problem: the system choosing the wrong pre-approved message for the classification it produced. This left the wording, tone and sensory aspects to the other hazard owner. Another challenge was keeping the references aligned as the SRS changed. Whenever we clarified a lesson rule or added a safety requirement, I had to check the FMEA again rather than assuming the old wording still worked. It took extra revisions, but the result was more precise and easier to review.

**Moustafa:** The main pain point was that some of my hazards depended on requirements nobody had written yet. For example, the hazards about refinement changing a drawing too much and progress views misleading adults belong with Muhammad's creative-drawing and progress-summary requirements, which weren't written when I was doing my rows. I couldn't just leave the mitigation column empty, so where a goal or business rule already covered the hazard, I traced to that, and where nothing did, I wrote my own safety requirements, SAF-MOU-001 and SAF-MOU-002.

It also meant my rows came in late, because I needed my own usability sections done first so the hazards had something to point to.

3. Which of your listed risks had your team thought of before this deliverable, and which did you think of while doing this deliverable? For the latter ones (ones you thought of while doing the Hazard Analysis), how did they come about?

   **Rayan:** Before this deliverable, the team had already thought of two of my risks. Because Squigglr runs in a browser, we expected that attempts could be lost or not saved on an unreliable browser or device (H-RAY-002). We also knew that adults might read a child's records as a diagnosis (H-RAY-003), which is why the problem statement already avoids any diagnostic claims.

   The other five risks I only thought of while doing the hazard analysis. They came about because this was the first time I had to work out the requirements for the application in detail, while writing the SRS alongside this document. Most of them came from the critical assumptions, and several of those assumptions came from our interview with the specialists on October 1, 2026. The specialists said children can use the app through images and icons while adults handle setup, which led to the assumption that an adult sets up each session. Asking what happens if no adult is actually watching led to the risk of a child reaching adult-only actions (H-RAY-001), and, together with defining what happens after repeated failed attempts, to the risks of a child becoming distressed (H-RAY-004) and of a lesson being restarted straight after it ends (H-RAY-005). The specialists also said Squigglr may be used at home, at school or in other settings, which led to the assumption that a device may be shared and to the risk of a child's records being saved under the wrong profile (H-RAY-006). The last risk came from our meeting on October 14, 2026, where Dr. Anand advised that adults should set the thresholds for each child; this raised the risk of a threshold being set to a value that does not suit the child (H-RAY-007).

   Looking back, most of my risks only became visible once the requirements were concrete enough to question. The risks we knew about beforehand were general ones that apply to almost any browser app, while the ones I found during this deliverable were specific to how children and adults will actually use Squigglr. This showed me that the hazard analysis cannot be done well in isolation; it depends on knowing the requirements and on hearing from people who understand the users.

   **Shamil:** We thought of hazards related to unauthorized account access (H-SHA-001) before the hazard analysis simply because it was obvious and a basic need to be addressed. Additionally, we had an idea of the issues that would come with using a third-party AI provider, including PII leakage to the provider (H-SHA-002), provider outage (H-SHA-004), and inappropriate/inaccurate output (H-SHA-005) which are also obvious.

The hazard of false save confirmation (H-SHA-003)c ame to use later as we consulted more experts and realized the value of user trust, especially regarding transparency for parents of child users. We did not want to mislead parents or have them find out too late that their childs progress was not saved; after brainstorming, we chose to tackle this by making the saving confirmation more robust.

**Ahmed:** Before we started this deliverable, I had already thought about the possibility of drawing input being incomplete and of an incorrect attempt being allowed to pass. Those were concerns that came up naturally while defining how a guided lesson should evaluate strokes and move a learner to the next stage. They became H-AHM-001 and H-AHM-004 once I described their causes, effects and mitigations more systematically.

The FMEA made me consider several less obvious failures. For H-AHM-002, I started thinking about what happens when input events are reported more than once. Two overlapping strokes might be perfectly intentional, but one event being duplicated by the software is a different problem, and simply deleting repeated-looking lines would not solve it. H-AHM-003 came from looking at how the canvas changes size: even if the stylus reports valid points, a scaling or coordinate-mapping mistake could shift the drawing before it is scored. Those cases led to the new input-integrity safety requirements.

I also looked beyond the drawing itself. H-AHM-005 came from asking whether a correct classification could still produce the wrong feedback if the message lookup used the wrong stage or identifier. H-AHM-006 came from tracing the failure streak, improvement rule and adult-approval step together; an error there could suggest a stage change at the wrong time or apply one without permission. Finally, H-AHM-007 came from considering what happens when a learner's saved settings are changed or profiles are switched. That exercise showed me that a lot of meaningful risks are not obvious until you follow the data and decisions across the entire workflow.

**Moustafa:** Most of my risks were things we had already thought about before this deliverable, because we planned against them in the presentation we gave the specialists on October 1. We had already decided that feedback would highlight the unfinished part of a drawing instead of failing the attempt or finishing it for the child, that there would be no scores and fewer attempts wouldn't mean better skill, and that refinement would only tidy the child's own drawing, with the original always restorable. Those became H-MOU-002, H-MOU-004, H-MOU-005 and H-MOU-006. The other two came up while doing this deliverable. Sensory overload (H-MOU-001) came from the questionnaire, where one specialist warned about excessive visual or auditory stimuli, and the flashing part came from WCAG while I was writing the accessibility requirements. Confusing feedback (H-MOU-003) came from the same work: once I let adults mute audio and said feedback can't rely on colour alone, I started asking

what happens to a child who gets sound-only or colour-only feedback, or feedback they need to read.

4. Other than the risk of physical harm (some projects may not have any appreciable risks of this form), list at least 2 other types of risk in software products. Why are they important to consider?

   **Rayan:** Most software carries risks that have nothing to do with physical harm. One is privacy and security. Almost every application stores some information about its users, and if that information is exposed, accessed by the wrong person or mixed up between users, it can harm those users and destroy their trust in the product. This matters even more when the users are vulnerable. In Squigglr, for example, the records describe a child's difficulties, so they must only be visible to the adults responsible for that child. A second is data integrity and availability. People make decisions based on what software shows them, so data that is lost, corrupted or shown incorrectly can lead them to the wrong decision, even when nothing visibly crashes. In our case, an adult choosing a child's next lesson relies on the attempt records being complete and accurate. A third is psychological harm. Software that frustrates, confuses or misleads its users can affect their well-being or lead them to wrong conclusions, for example by implying something about a person that the software cannot actually judge. For Squigglr, that means avoiding repeated failure without a break and making sure progress summaries are never read as a diagnosis. These risks are important to consider because they are easy to overlook when there is no obvious physical danger, yet they can cause real harm to users and to the people who rely on the software.

   **Shamil:** Loss of privacy and trust in the application are two types of risks other than physical harm that are important to address. Having one's PII leaked can cause emotional distress and fear for safety. It is also irreversible once a malicious actor has access or if the information becomes public. Furthermore, it leads to the second harm which is loss of trust. This harms the products reputation and prevents potential customers from considering an otherwise helpful product.

   **Ahmed:** One important type is data integrity risk. Software can appear to work while quietly saving or interpreting the wrong information. That matters for Squigglr because a drawing could be captured incorrectly, marked as passing when it should not, or saved under the wrong result. Adults rely on those records to understand what happened during practice, so an inaccurate record is not just a technical bug; it can influence their next decision.

   Another type is privacy and security risk. A system may expose one person's information to someone who should not have access to it, or mix records between different users. Squigglr may be used on shared devices, and its records concern children, so we need to think about who can see each learner's history and how the active learner is identified.

Psychological or well-being risks also matter. Incorrect feedback, repeated failures or unexpected progression can confuse or frustrate a child even if there is no physical injury. These examples reminded me that software safety is about more than preventing a crash or a hardware failure. It also means making sure people can trust what the system tells them and that it behaves in a way appropriate for its users.

**Moustafa:** Two other types of risk are accessibility risk and legal or compliance risk. Accessibility risk is when some people simply can't use the software, and it's easy to miss because teams usually test on people like themselves. For Squigglr, a child who can't read yet, is colour-blind or has trouble with fine motor control could fail to use the app without anyone noticing, which is the opposite of what we're building it for. That's what most of my usability and accessibility requirements are there to prevent. Legal and compliance risk is when software breaks a law or regulation, which can get it pulled or fined even if it works fine. One of our specialists pointed out that WCAG is adopted as law in many countries, and Squigglr keeps records about children, so we have to think about both accessibility and privacy rules before we can put it in front of real families or schools.